

BFS Traversal

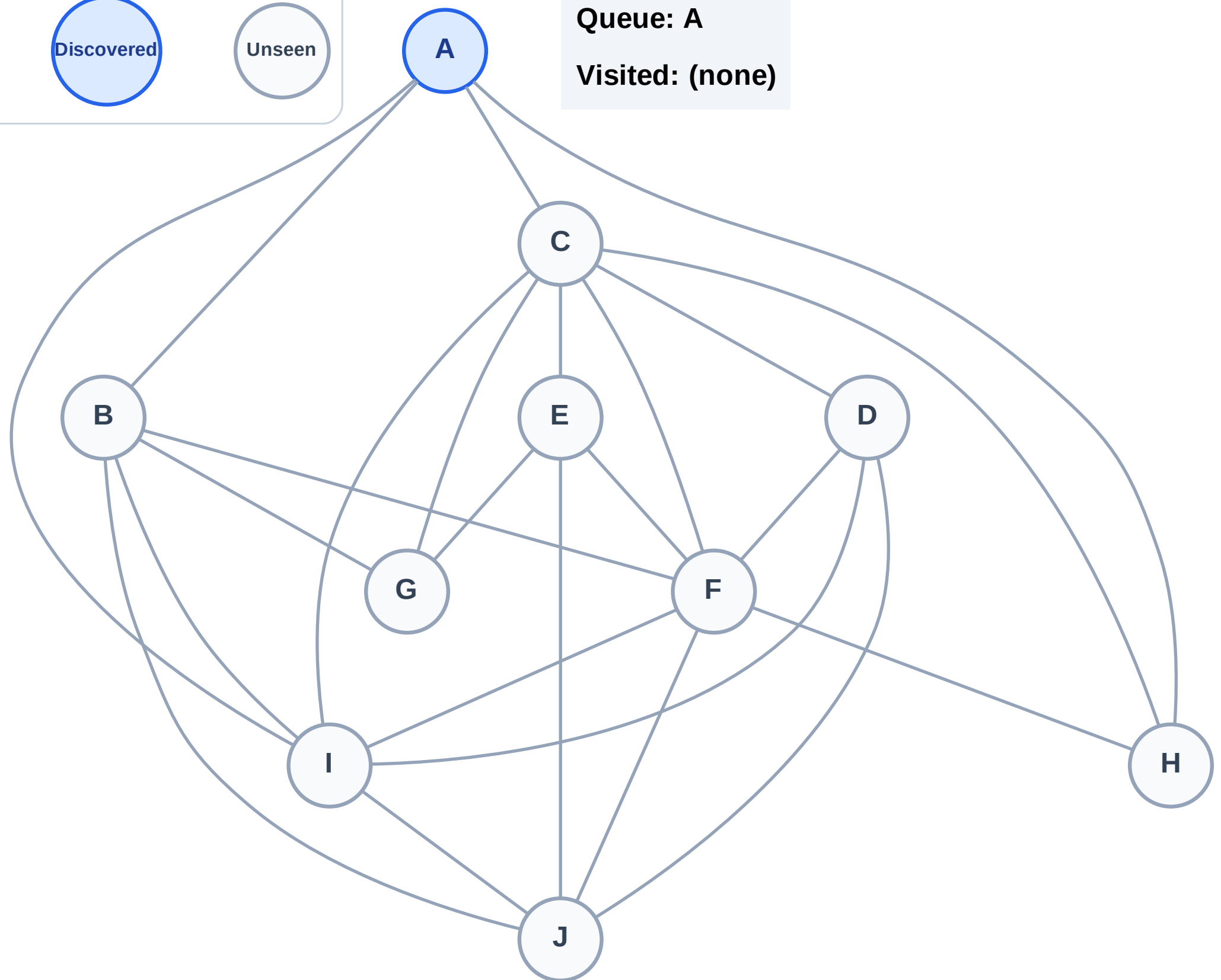
Step 1: Start at A

Legend



Queue: A

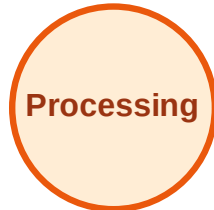
Visited: (none)



BFS Traversal

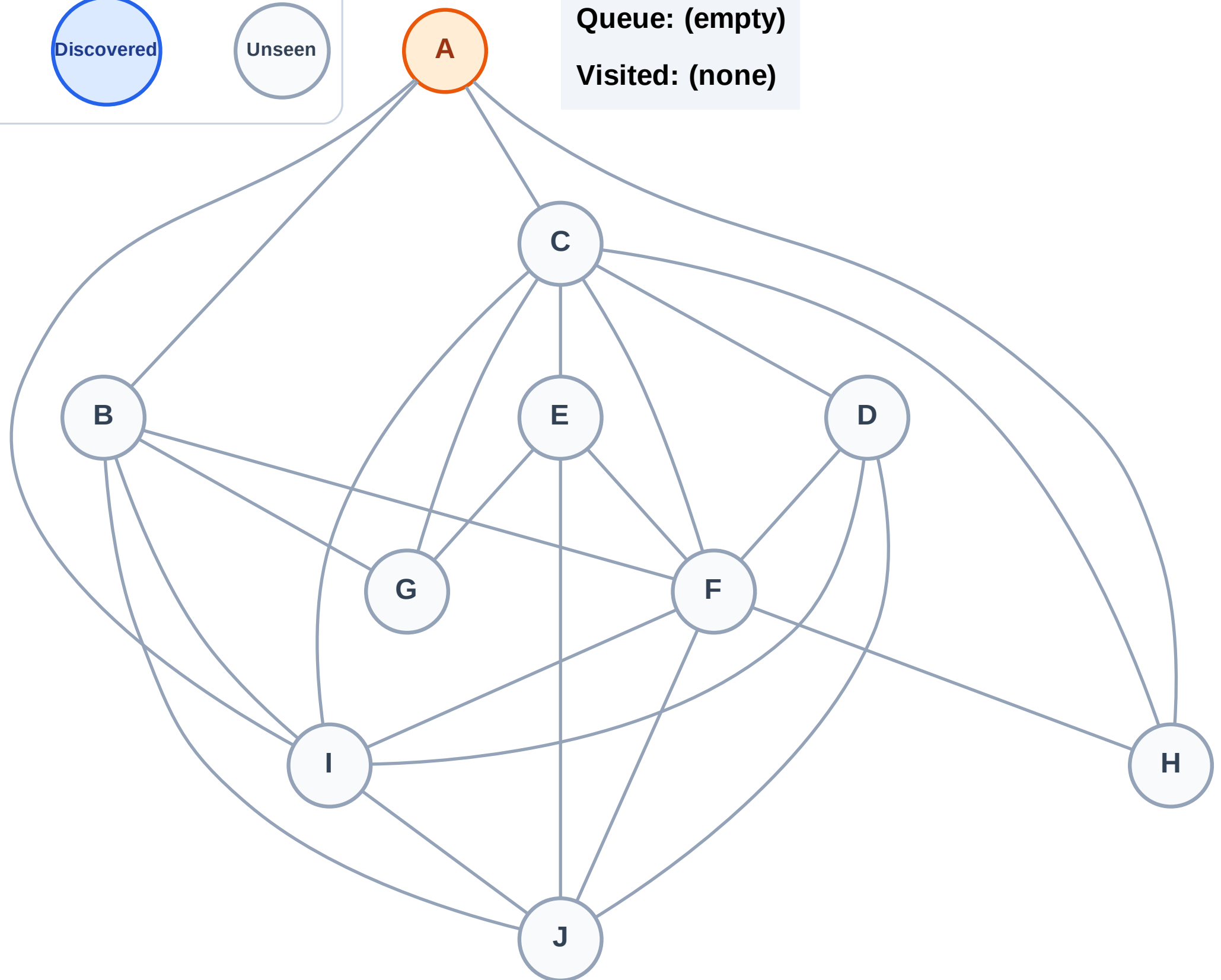
Step 2: Process node A

Legend



Queue: (empty)

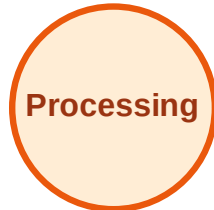
Visited: (none)



BFS Traversal

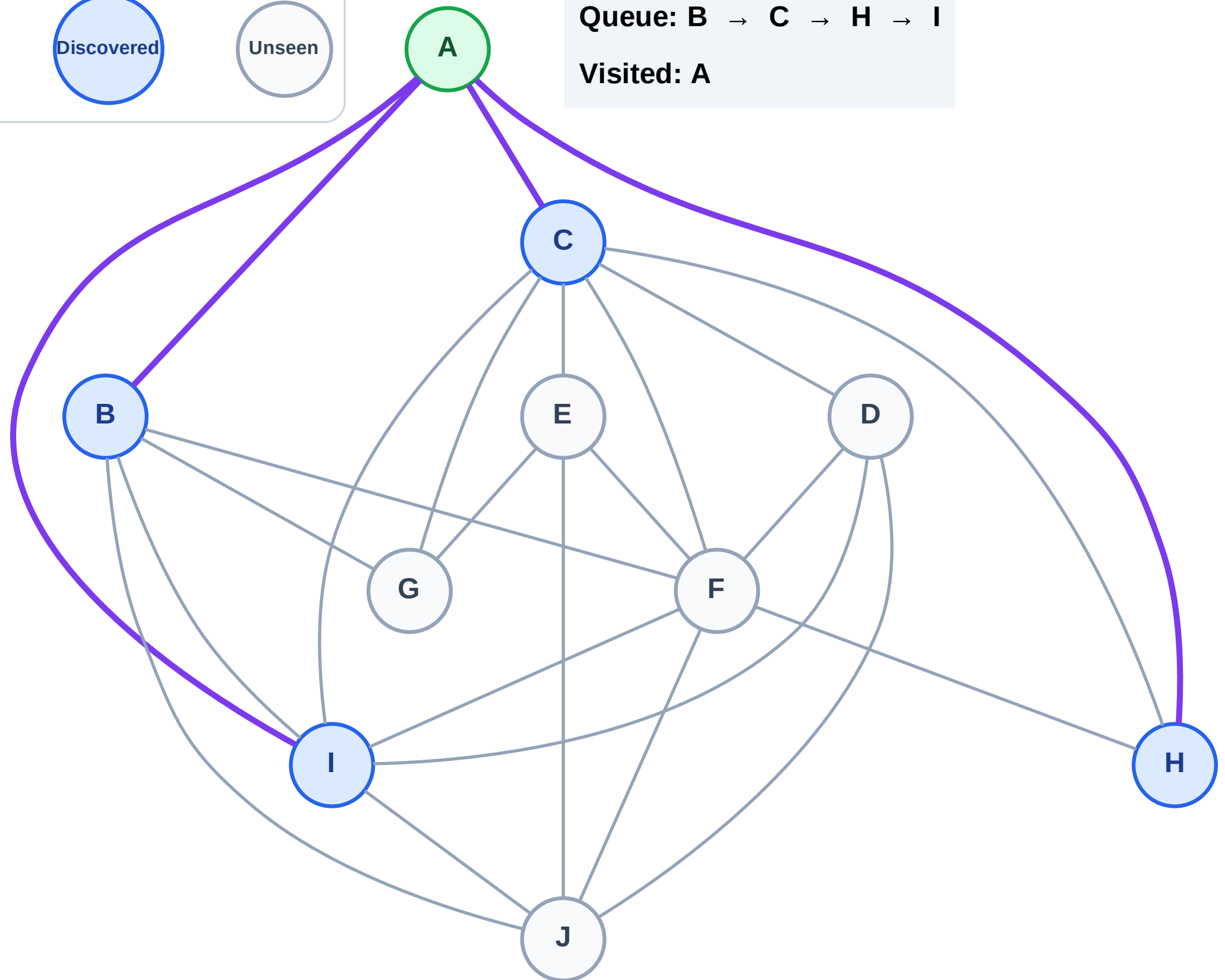
Step 3: Discover B, C, H, I from A

Legend



Queue: B → C → H → I

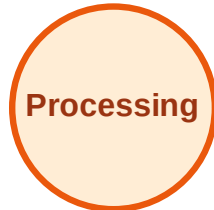
Visited: A



BFS Traversal

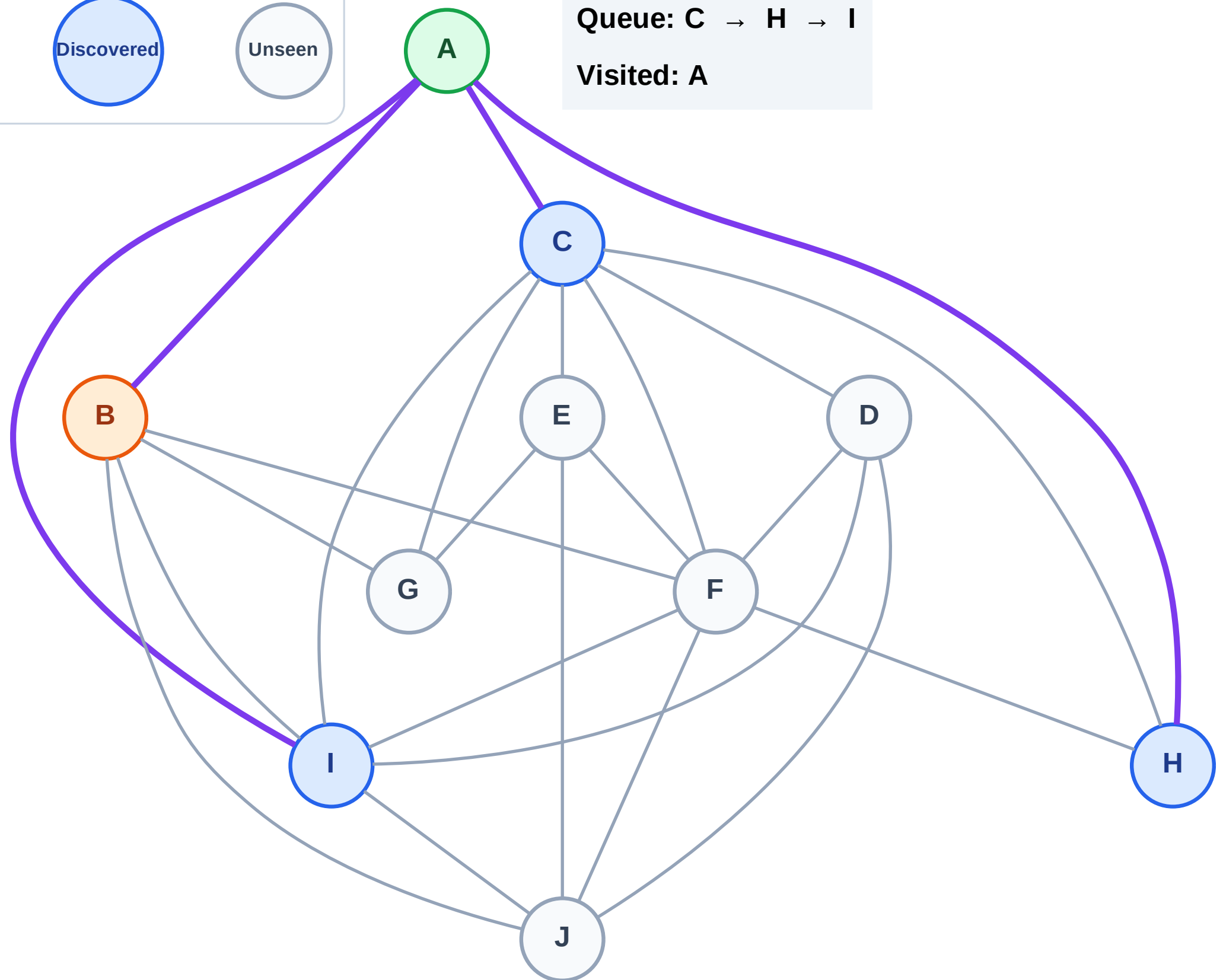
Step 4: Process node B

Legend



Queue: C → H → I

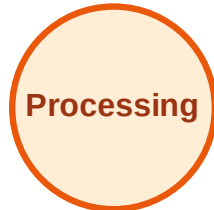
Visited: A



BFS Traversal

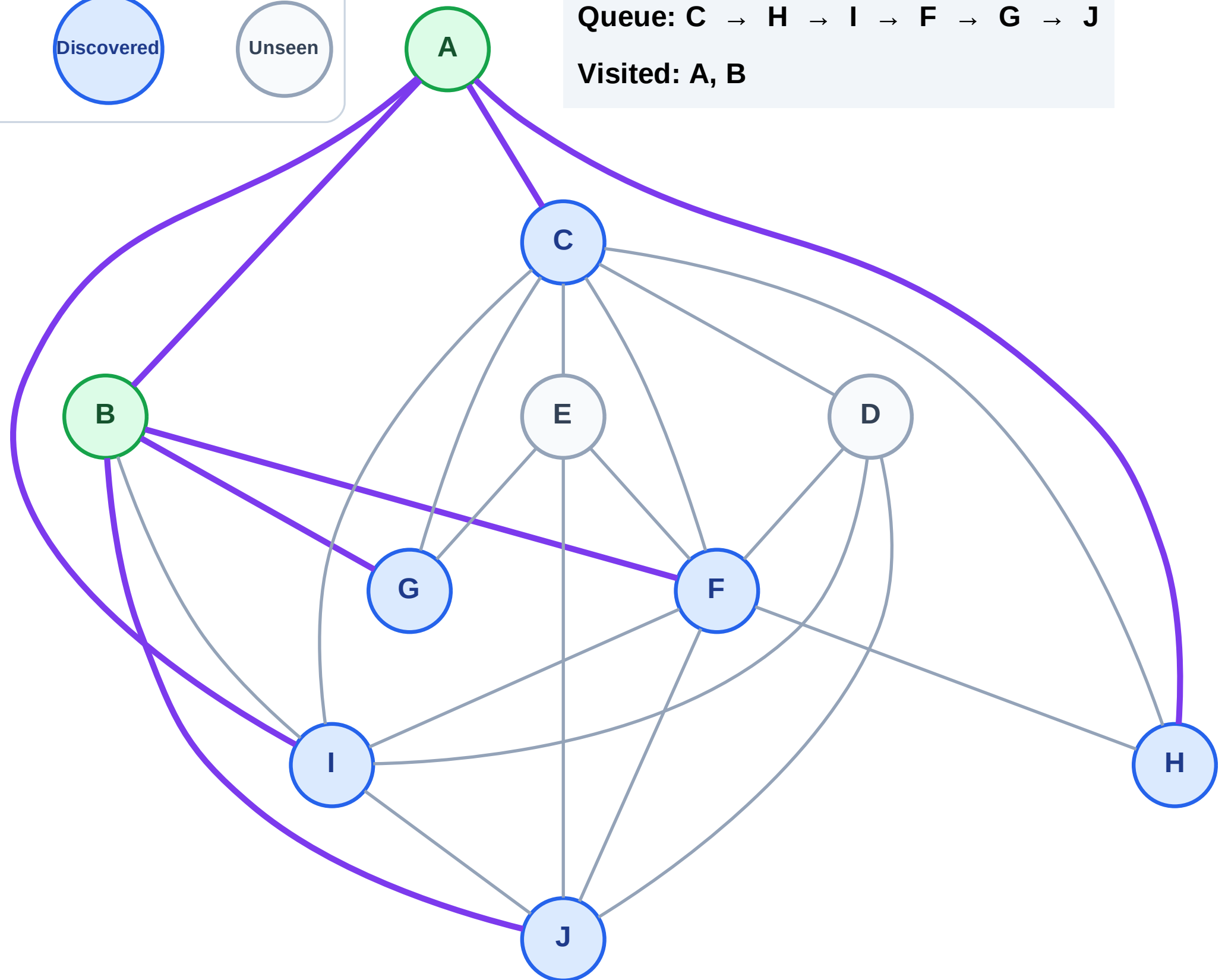
Step 5: Discover F, G, J from B

Legend



Queue: C → H → I → F → G → J

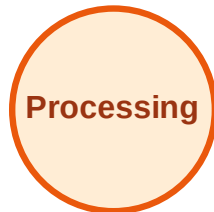
Visited: A, B



BFS Traversal

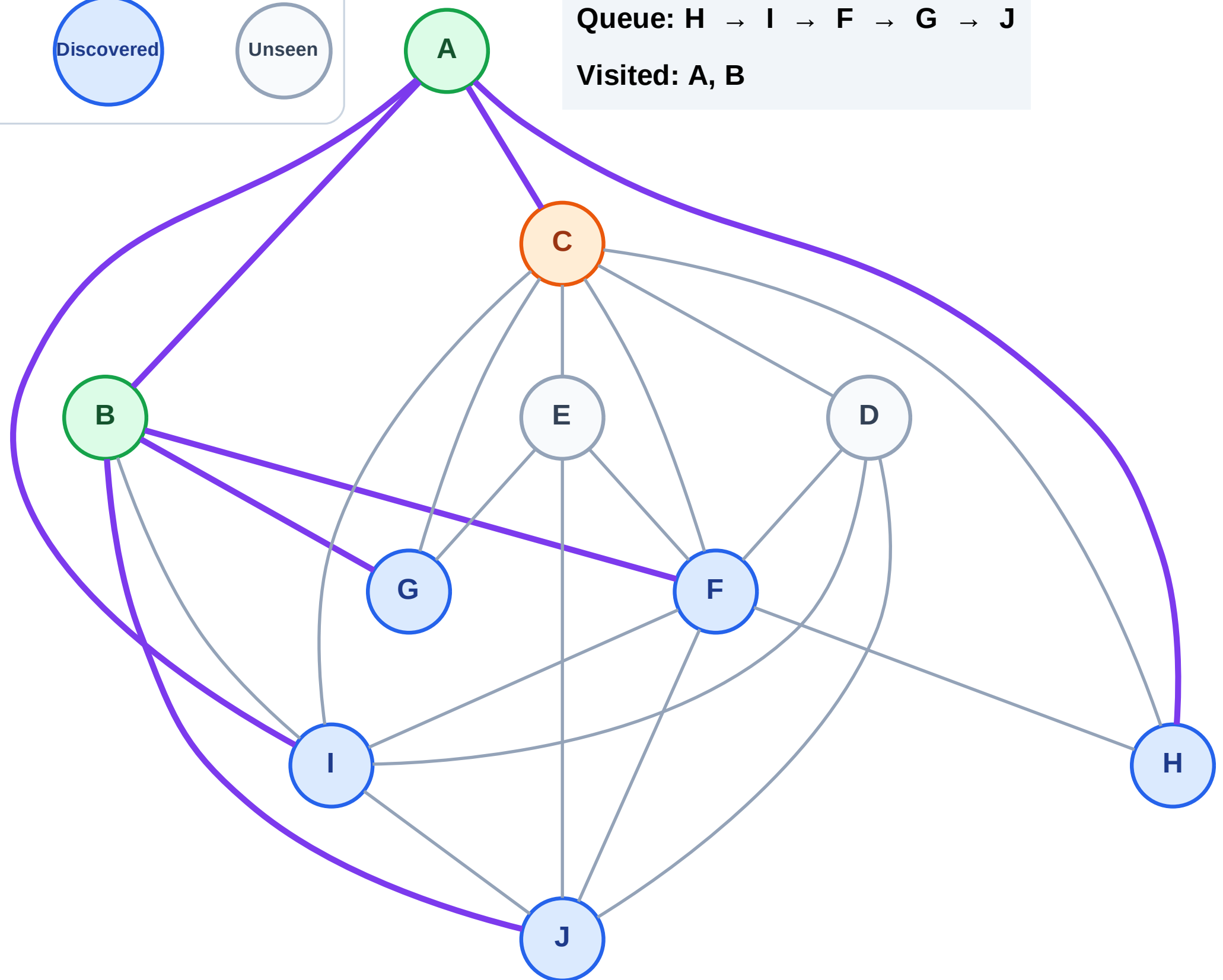
Step 6: Process node C

Legend



Queue: H → I → F → G → J

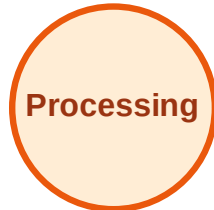
Visited: A, B



BFS Traversal

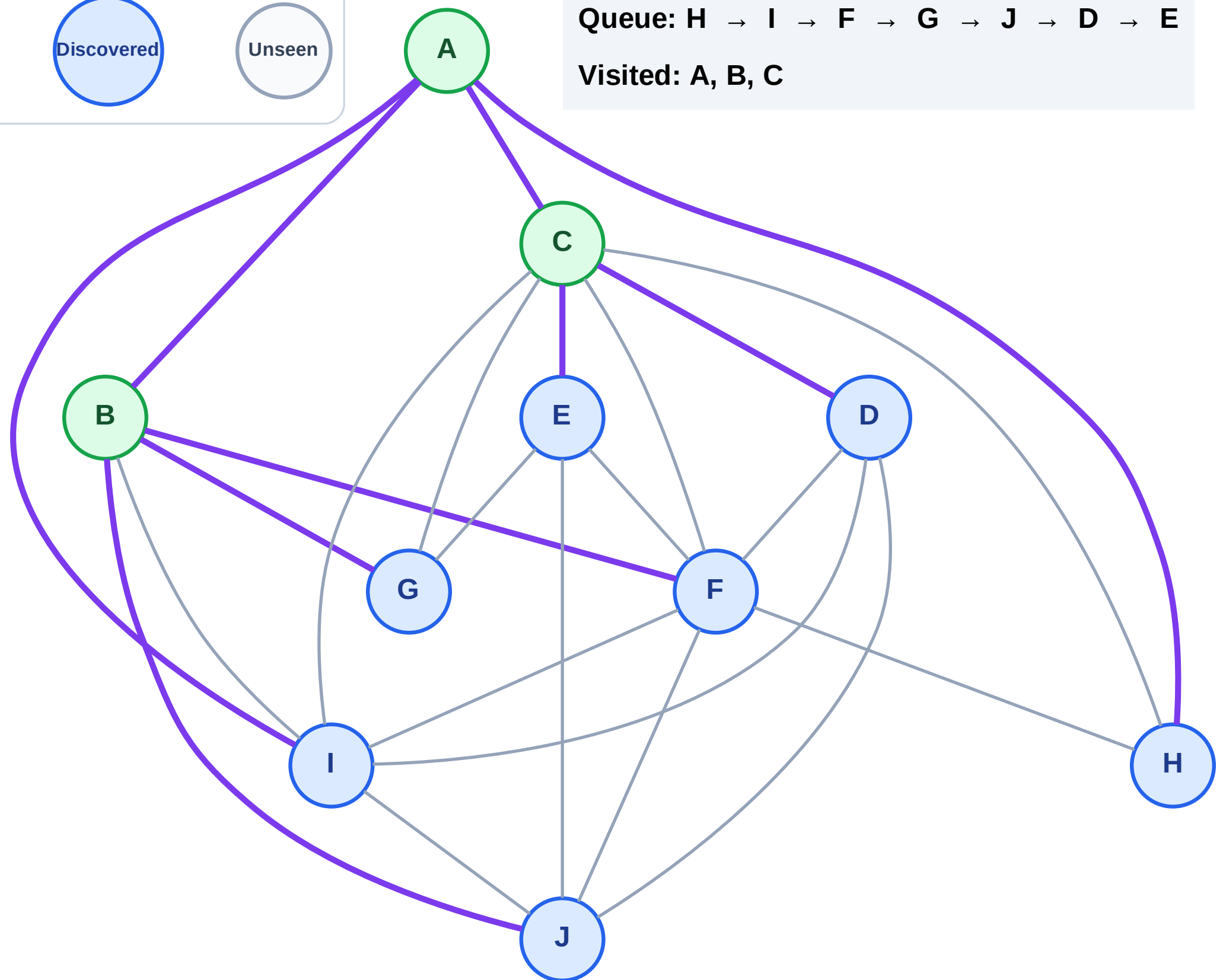
Step 7: Discover D, E from C

Legend



Queue: H → I → F → G → J → D → E

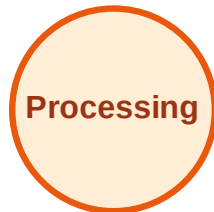
Visited: A, B, C



BFS Traversal

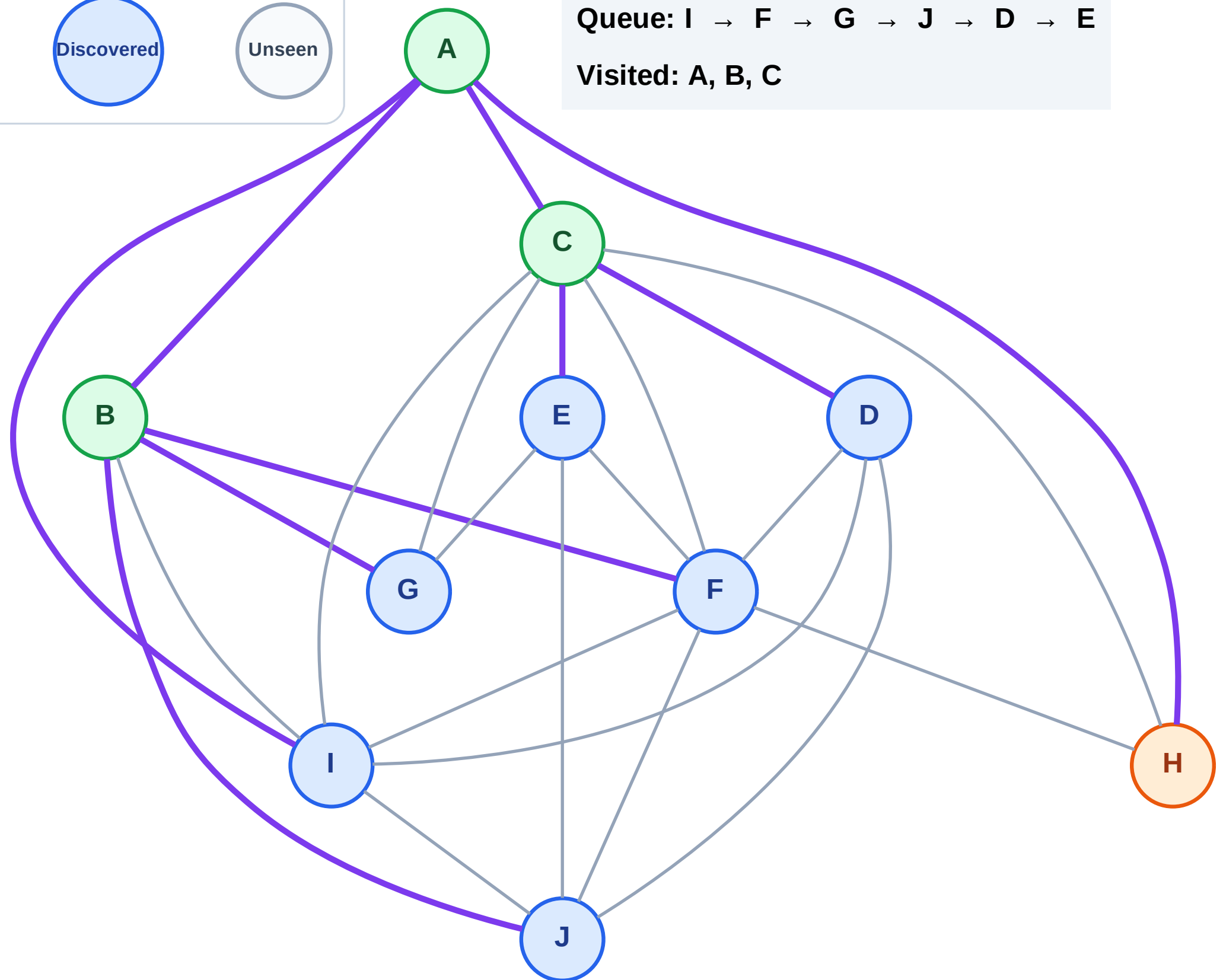
Step 8: Process node H

Legend



Queue: I → F → G → J → D → E

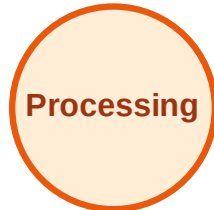
Visited: A, B, C



BFS Traversal

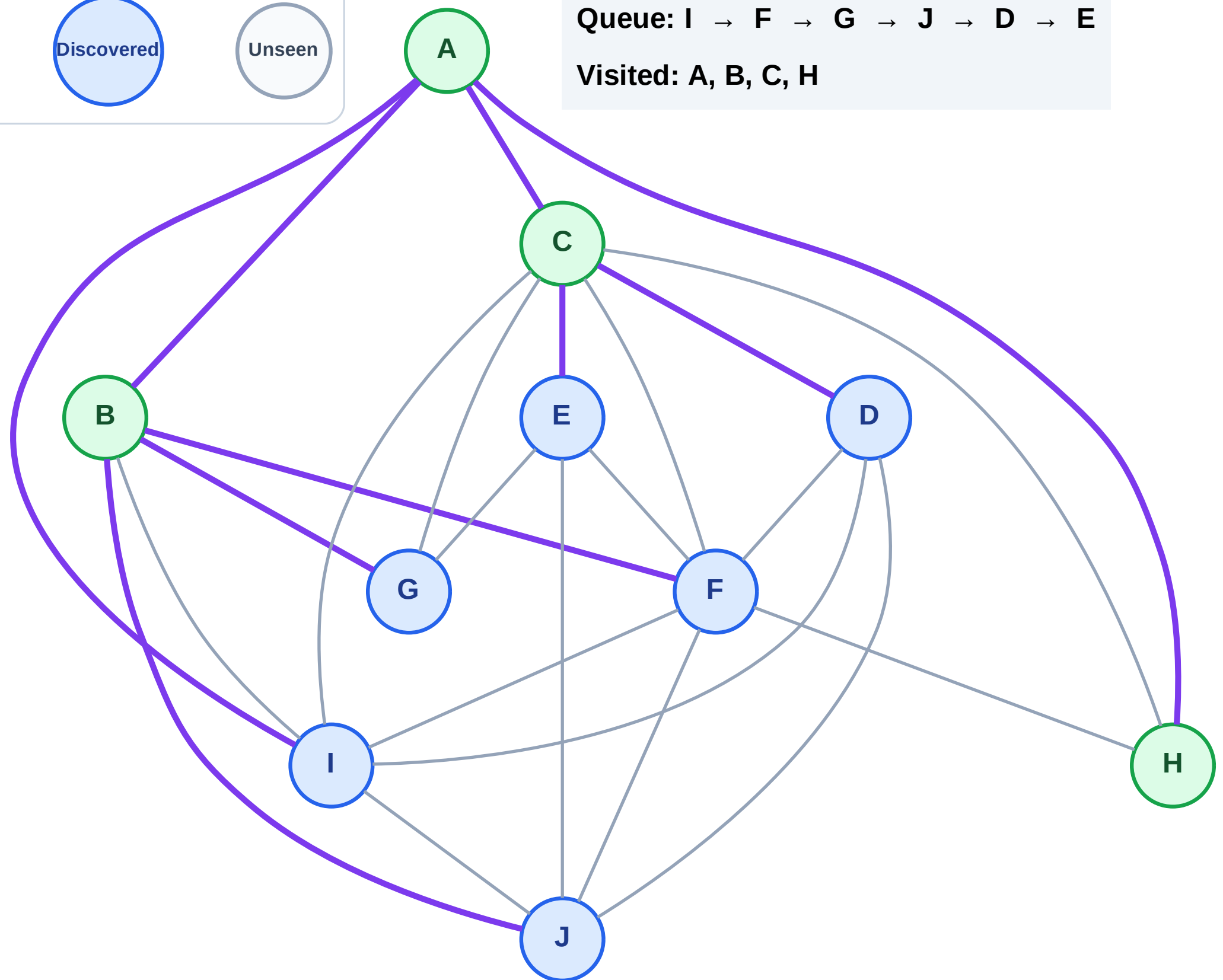
Step 9: No new neighbors from H

Legend



Queue: I → F → G → J → D → E

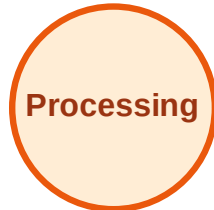
Visited: A, B, C, H



BFS Traversal

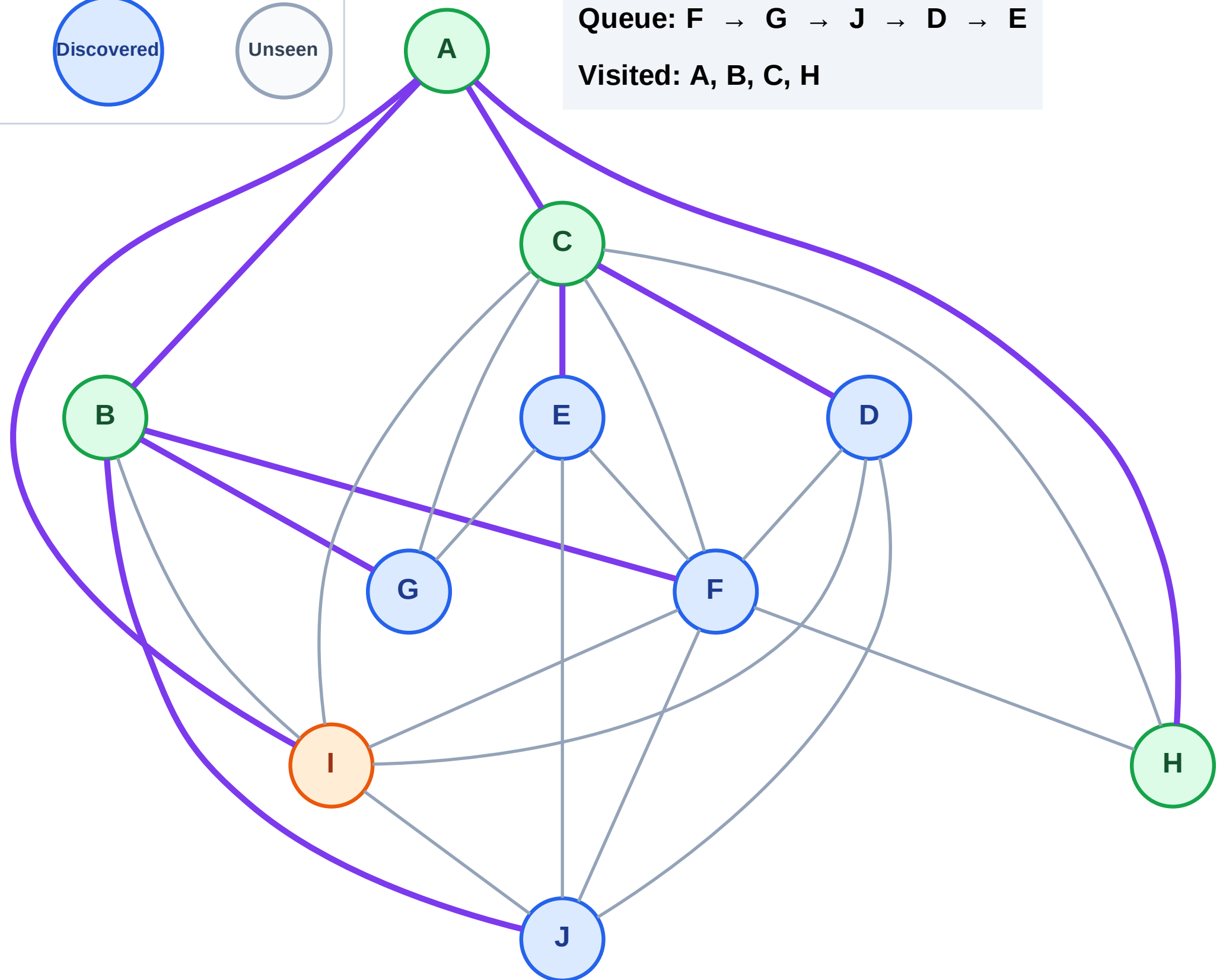
Step 10: Process node I

Legend



Queue: F → G → J → D → E

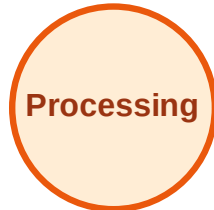
Visited: A, B, C, H



BFS Traversal

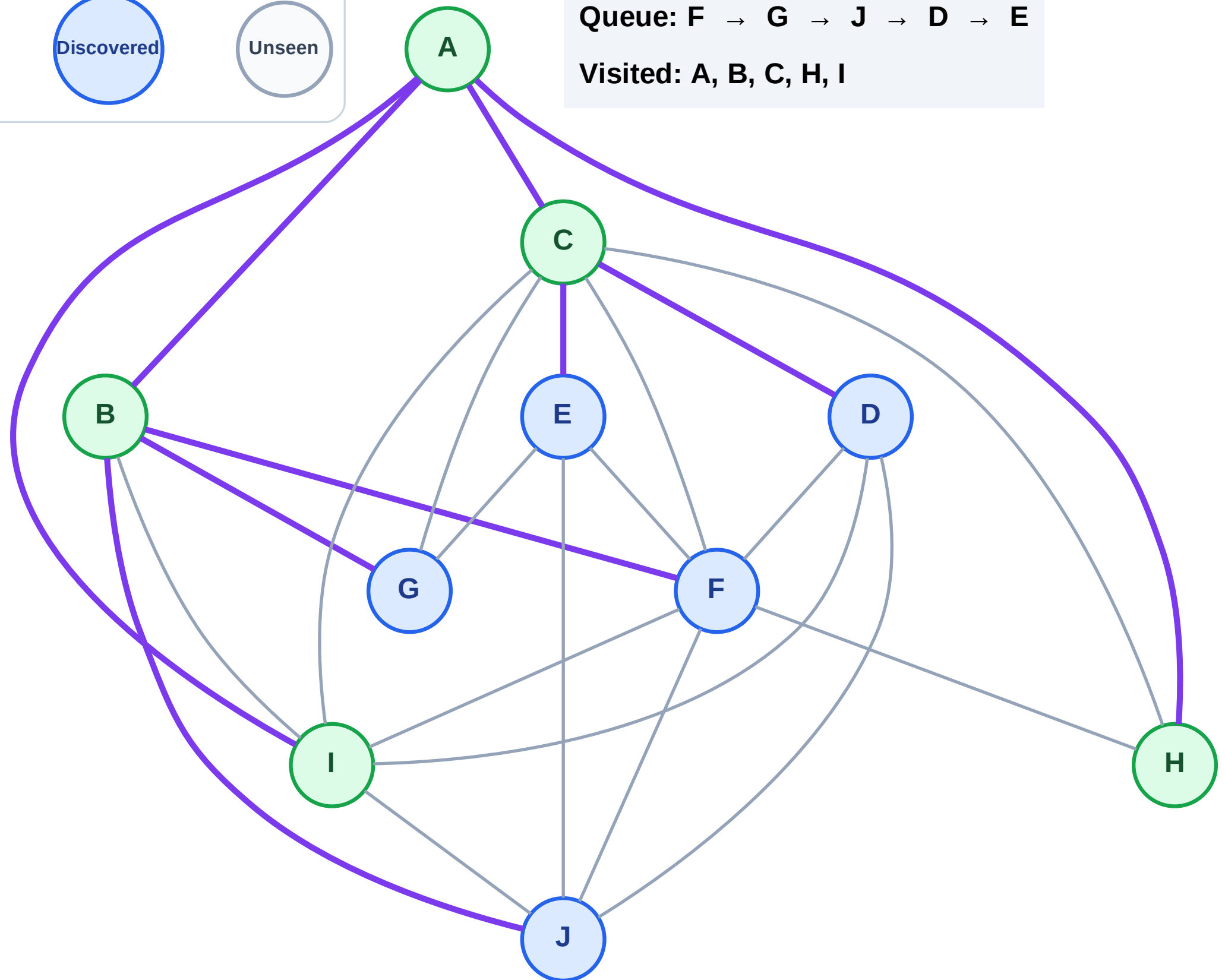
Step 11: No new neighbors from I

Legend



Queue: F → G → J → D → E

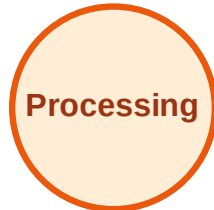
Visited: A, B, C, H, I



BFS Traversal

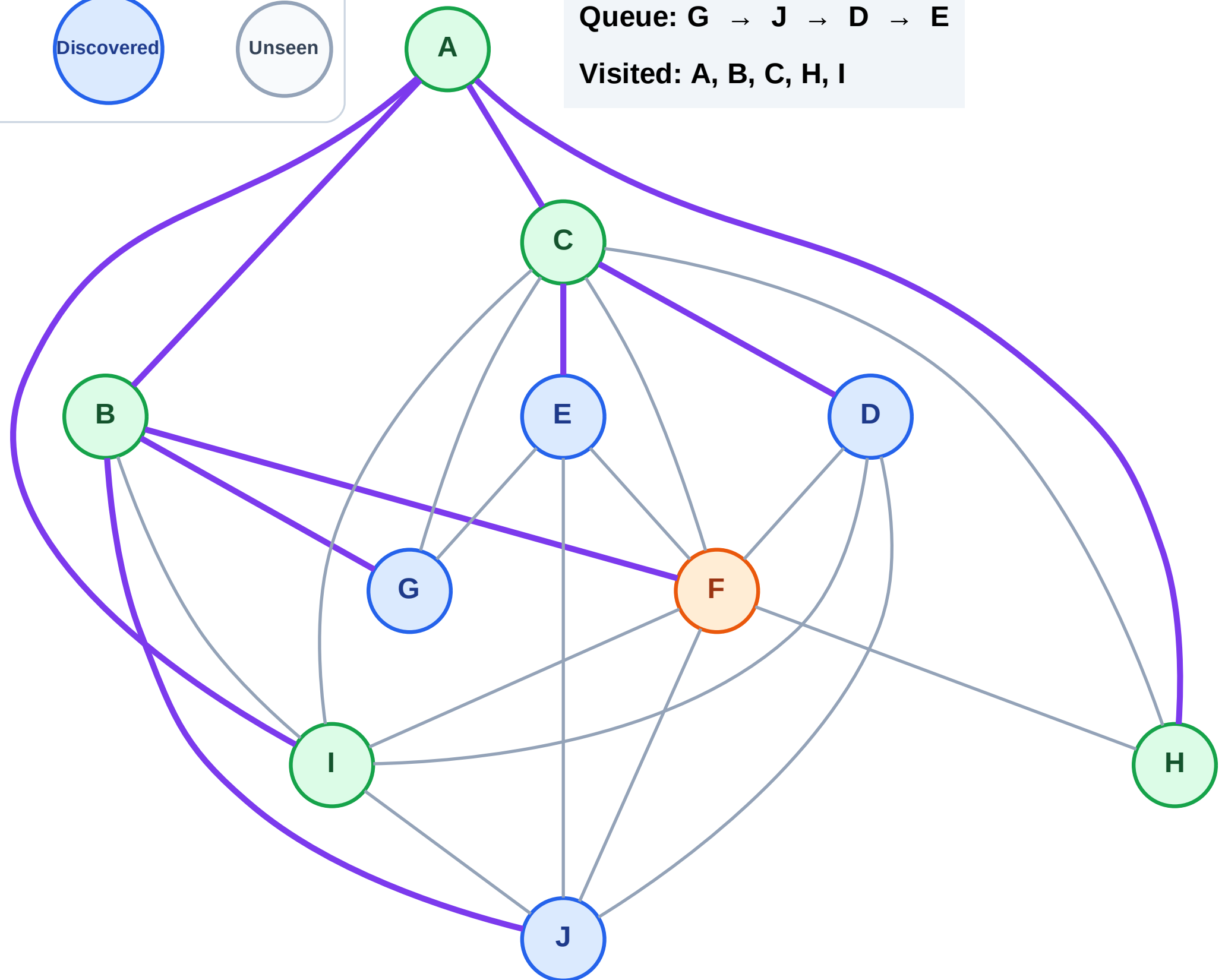
Step 12: Process node F

Legend



Queue: G → J → D → E

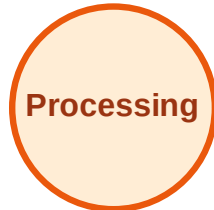
Visited: A, B, C, H, I



BFS Traversal

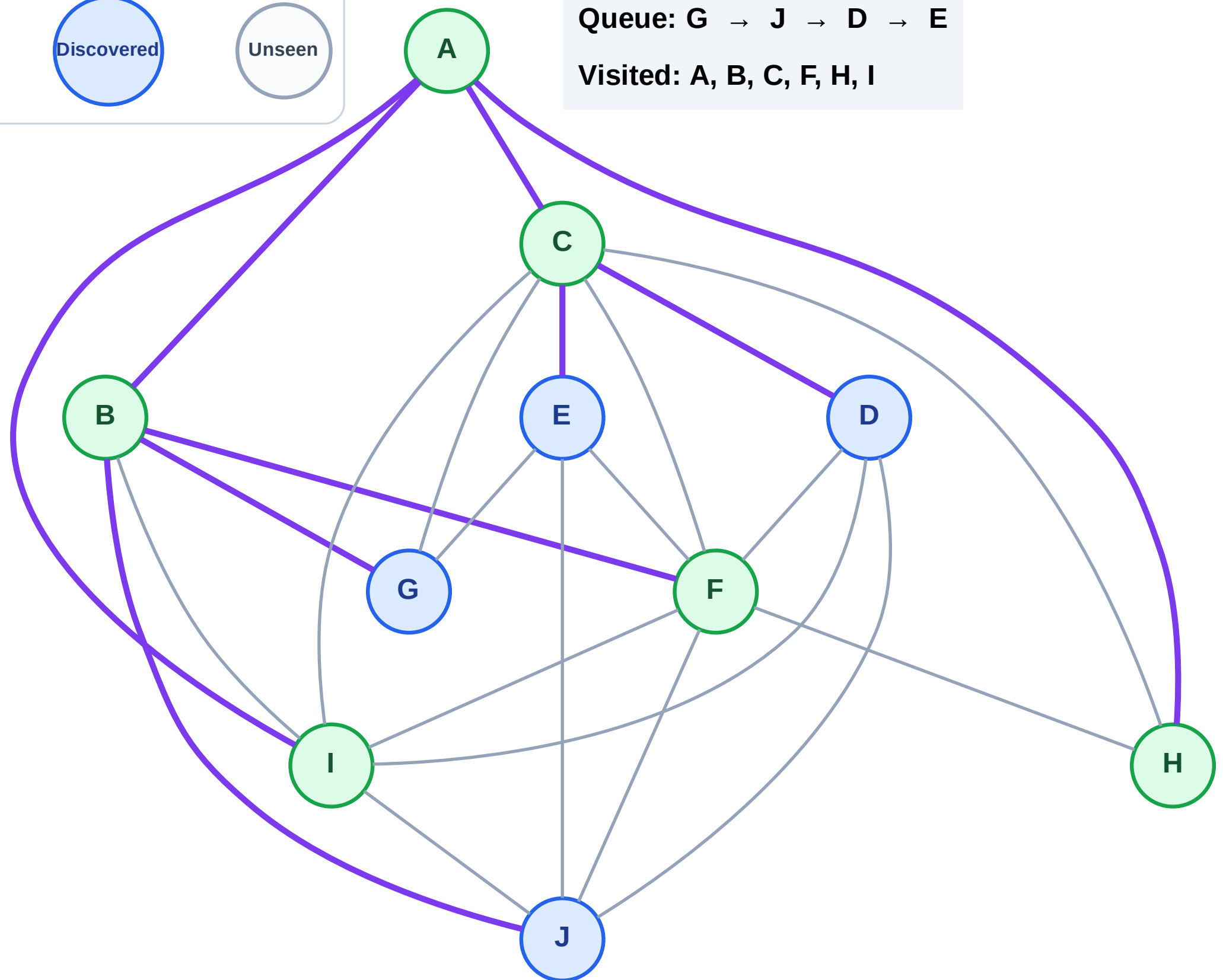
Step 13: No new neighbors from F

Legend



Queue: G → J → D → E

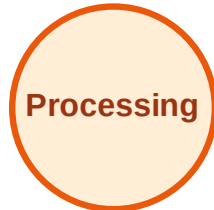
Visited: A, B, C, F, H, I



BFS Traversal

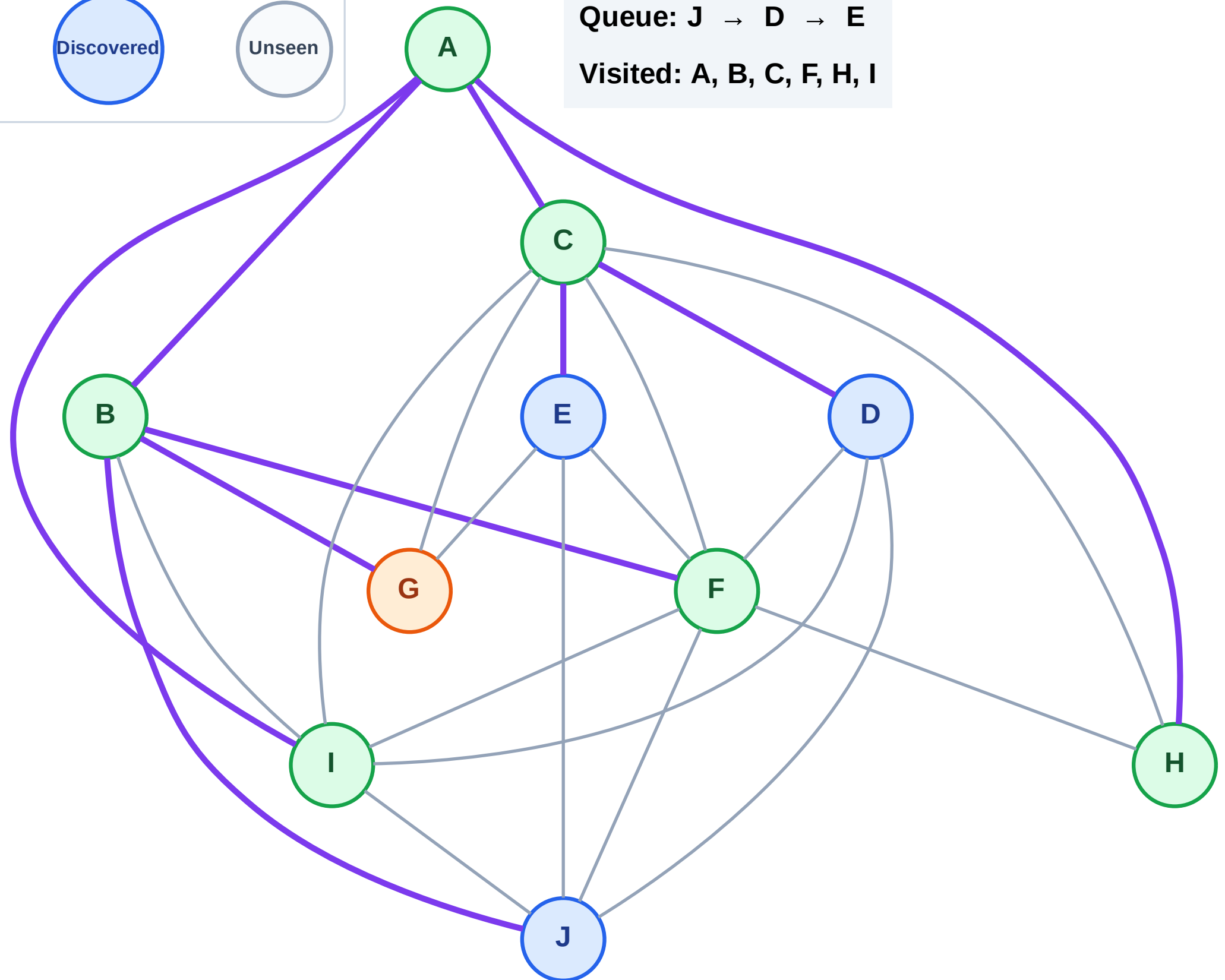
Step 14: Process node G

Legend



Queue: J → D → E

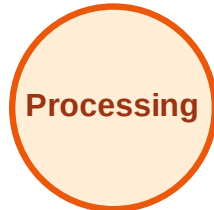
Visited: A, B, C, F, H, I



BFS Traversal

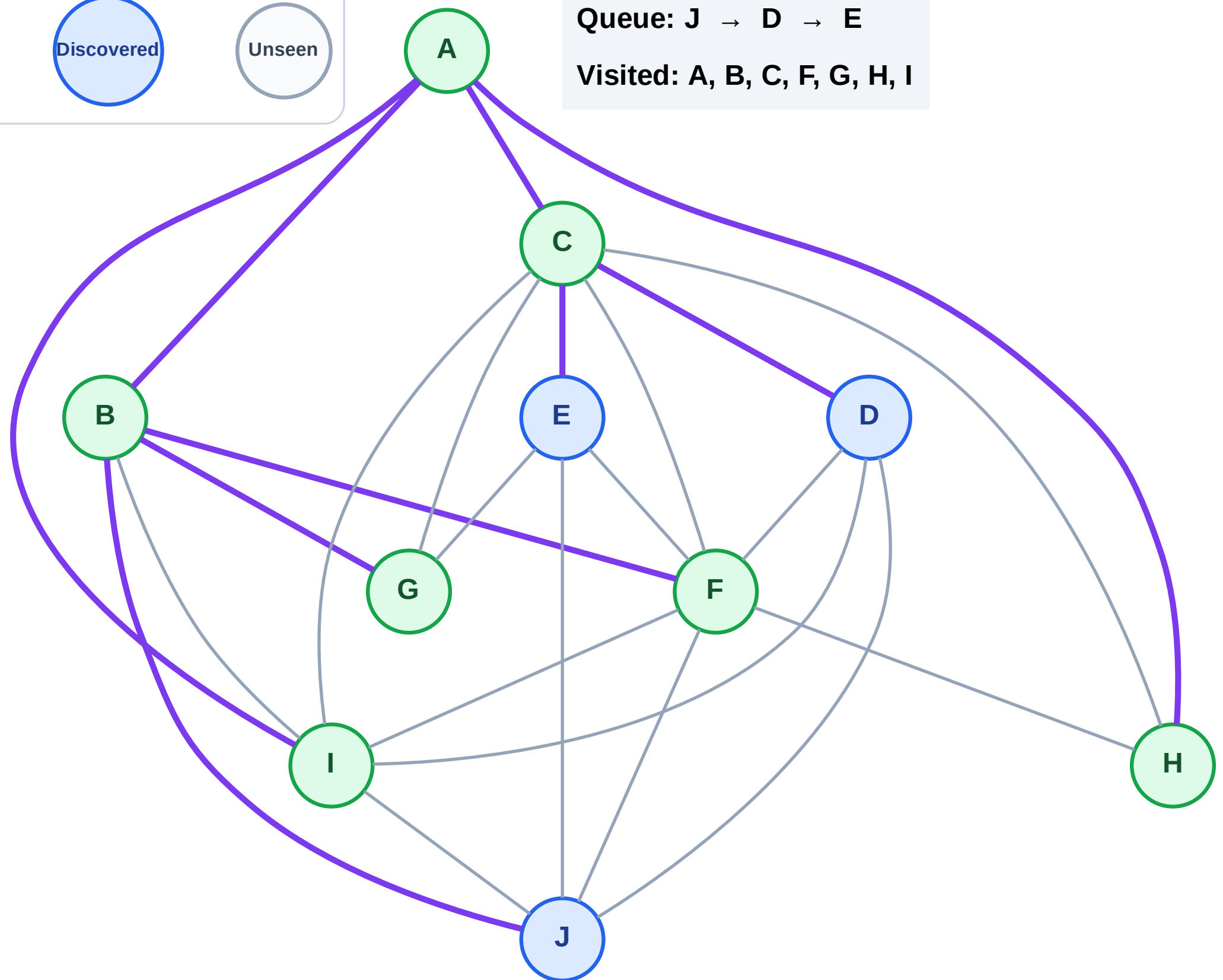
Step 15: No new neighbors from G

Legend



Queue: J → D → E

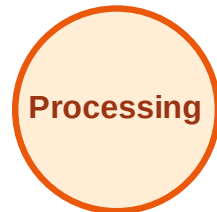
Visited: A, B, C, F, G, H, I



BFS Traversal

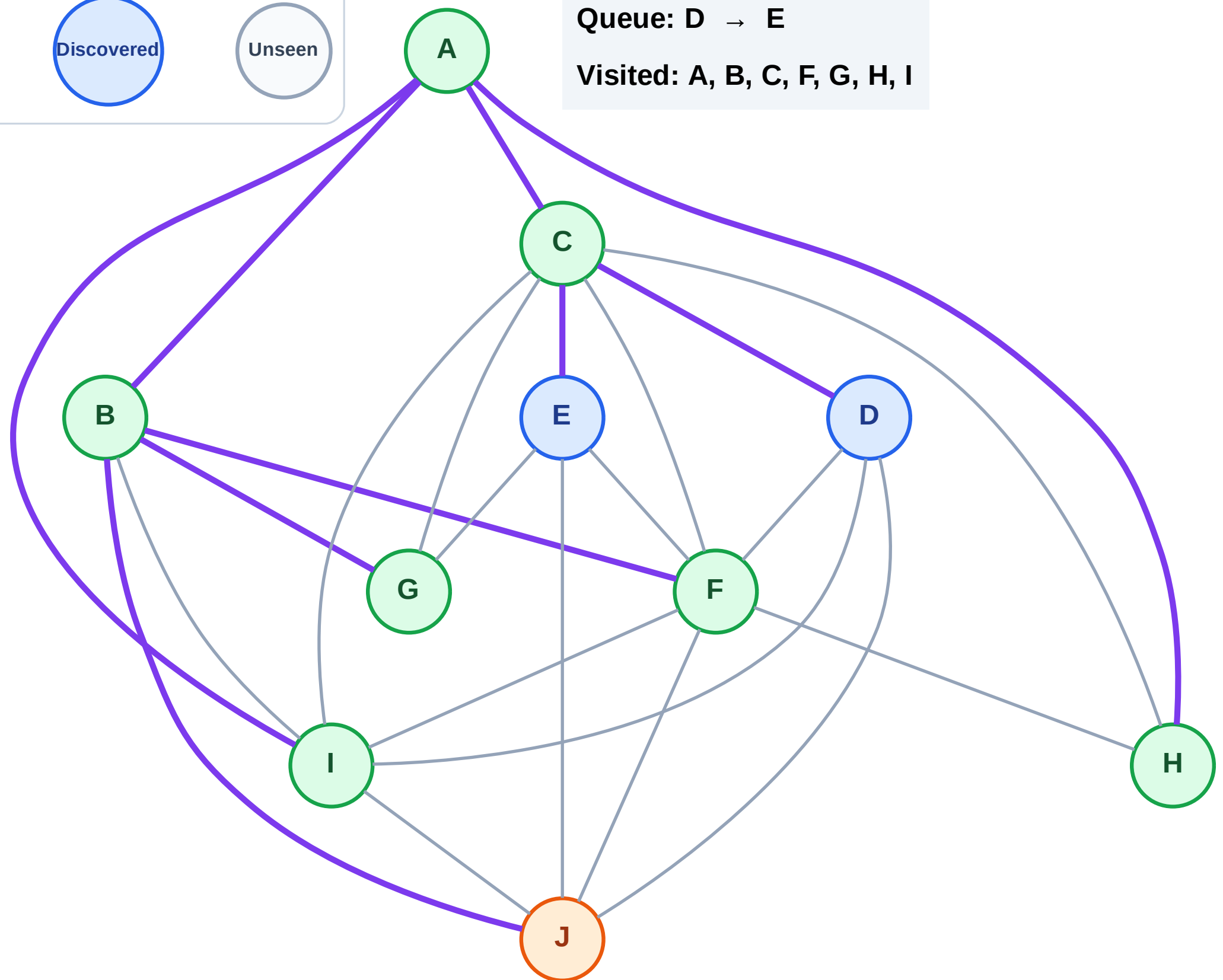
Step 16: Process node J

Legend



Queue: D → E

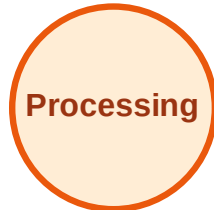
Visited: A, B, C, F, G, H, I



BFS Traversal

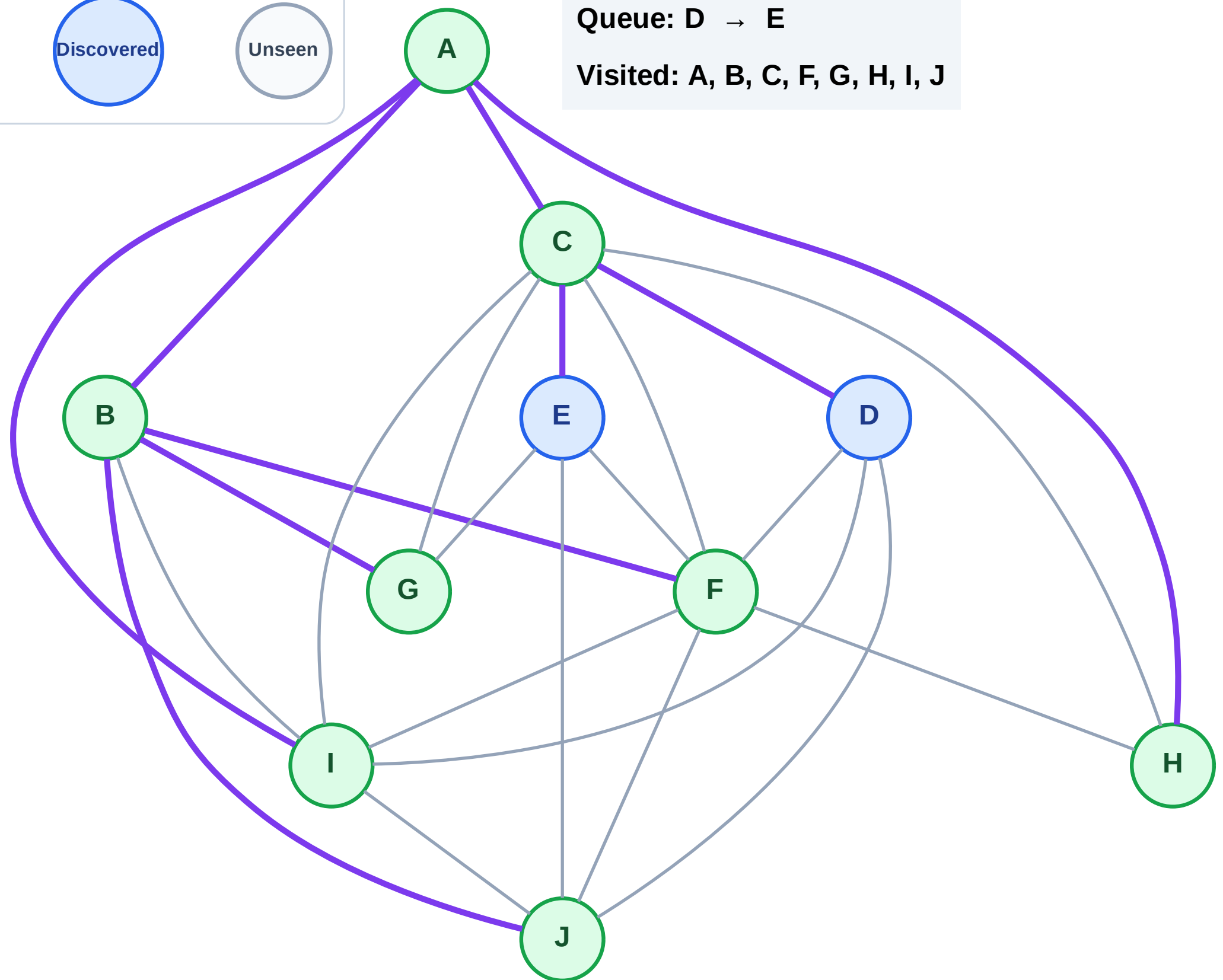
Step 17: No new neighbors from J

Legend



Queue: D → E

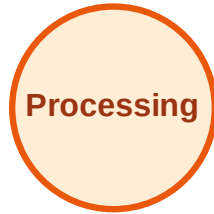
Visited: A, B, C, F, G, H, I, J



BFS Traversal

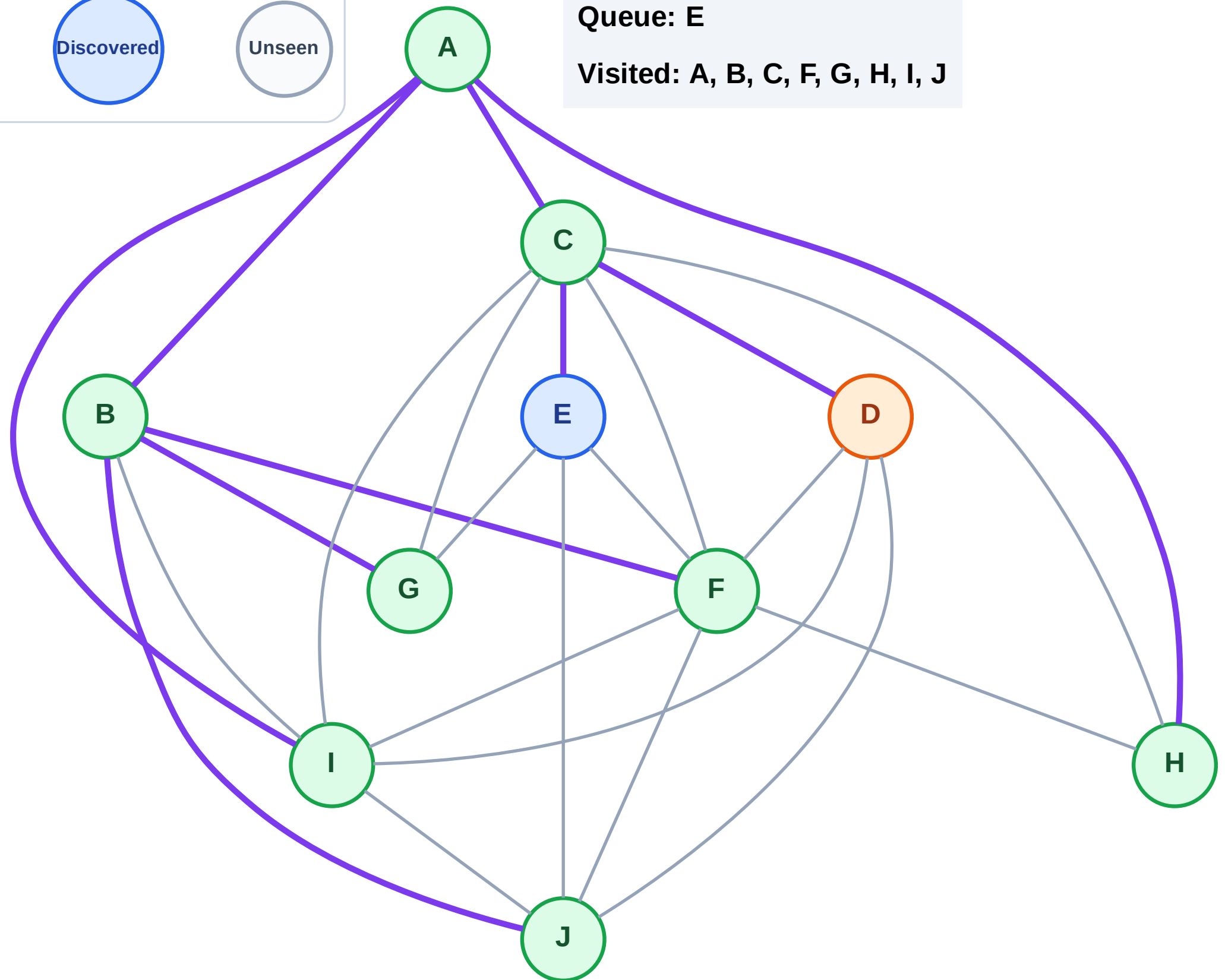
Step 18: Process node D

Legend



Queue: E

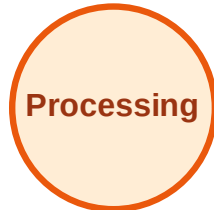
Visited: A, B, C, F, G, H, I, J



BFS Traversal

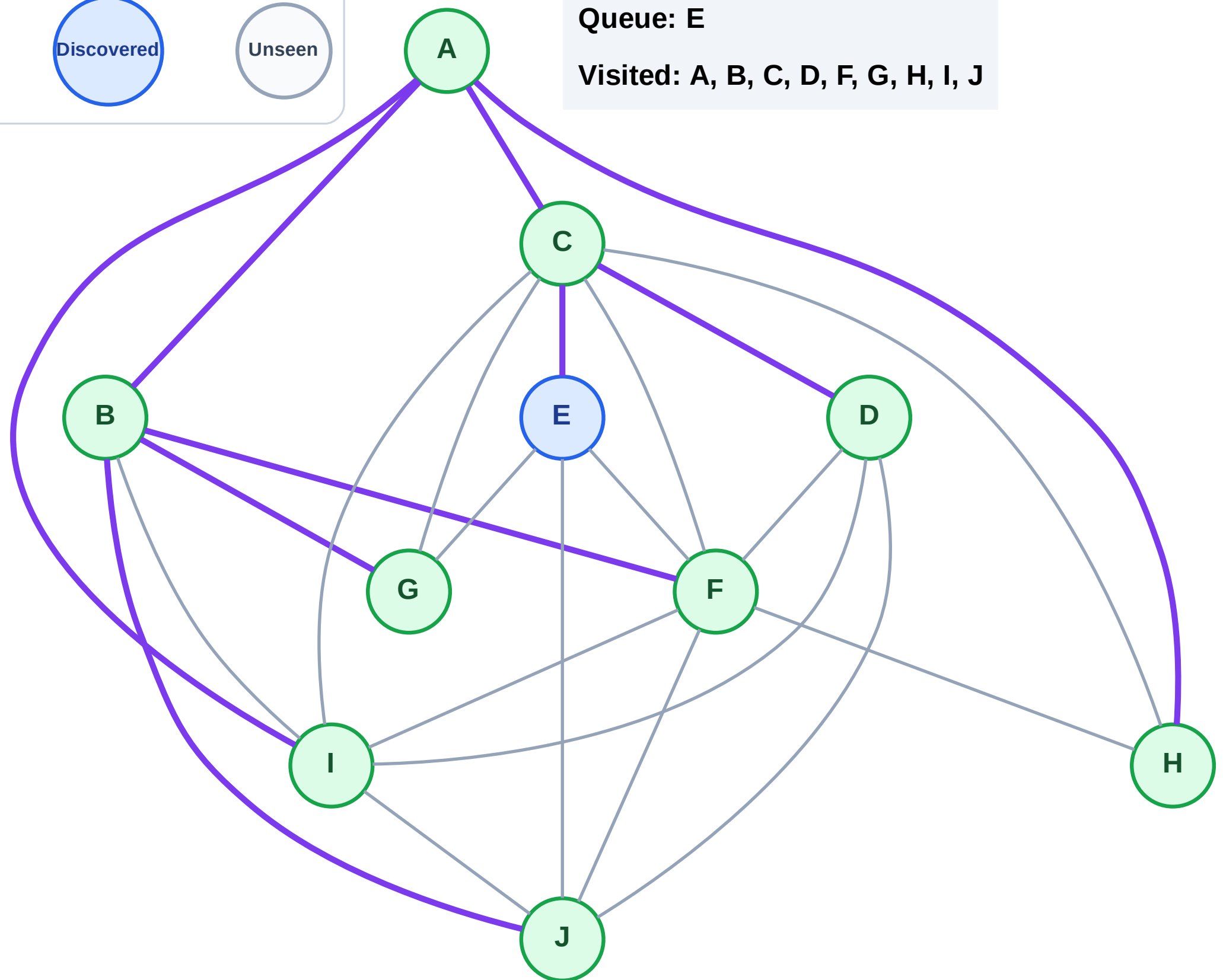
Step 19: No new neighbors from D

Legend



Queue: E

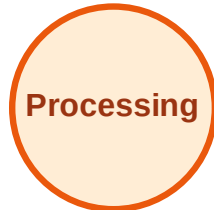
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

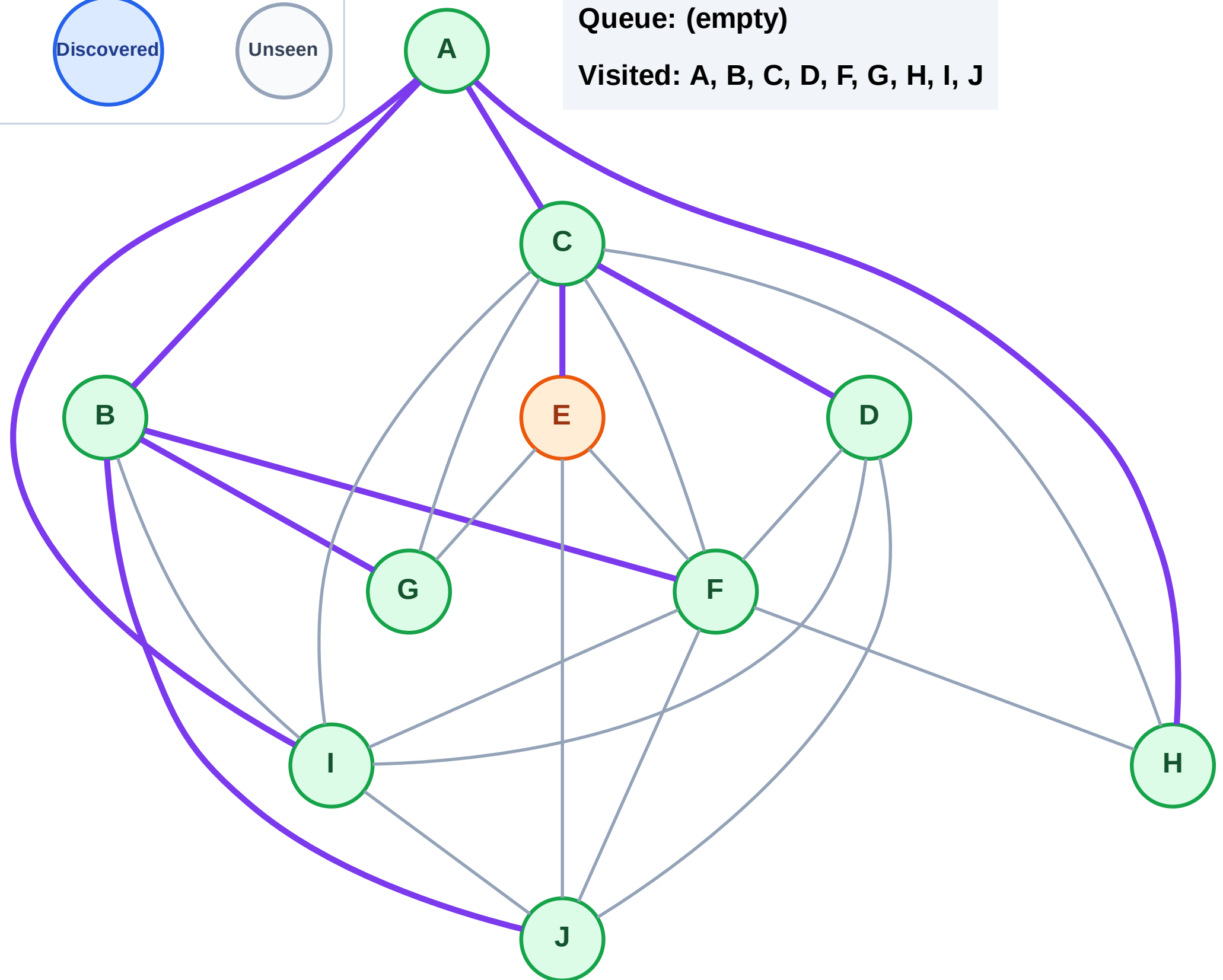
Step 20: Process node E

Legend



Queue: (empty)

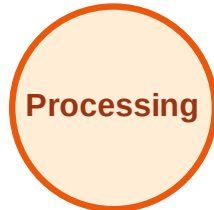
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

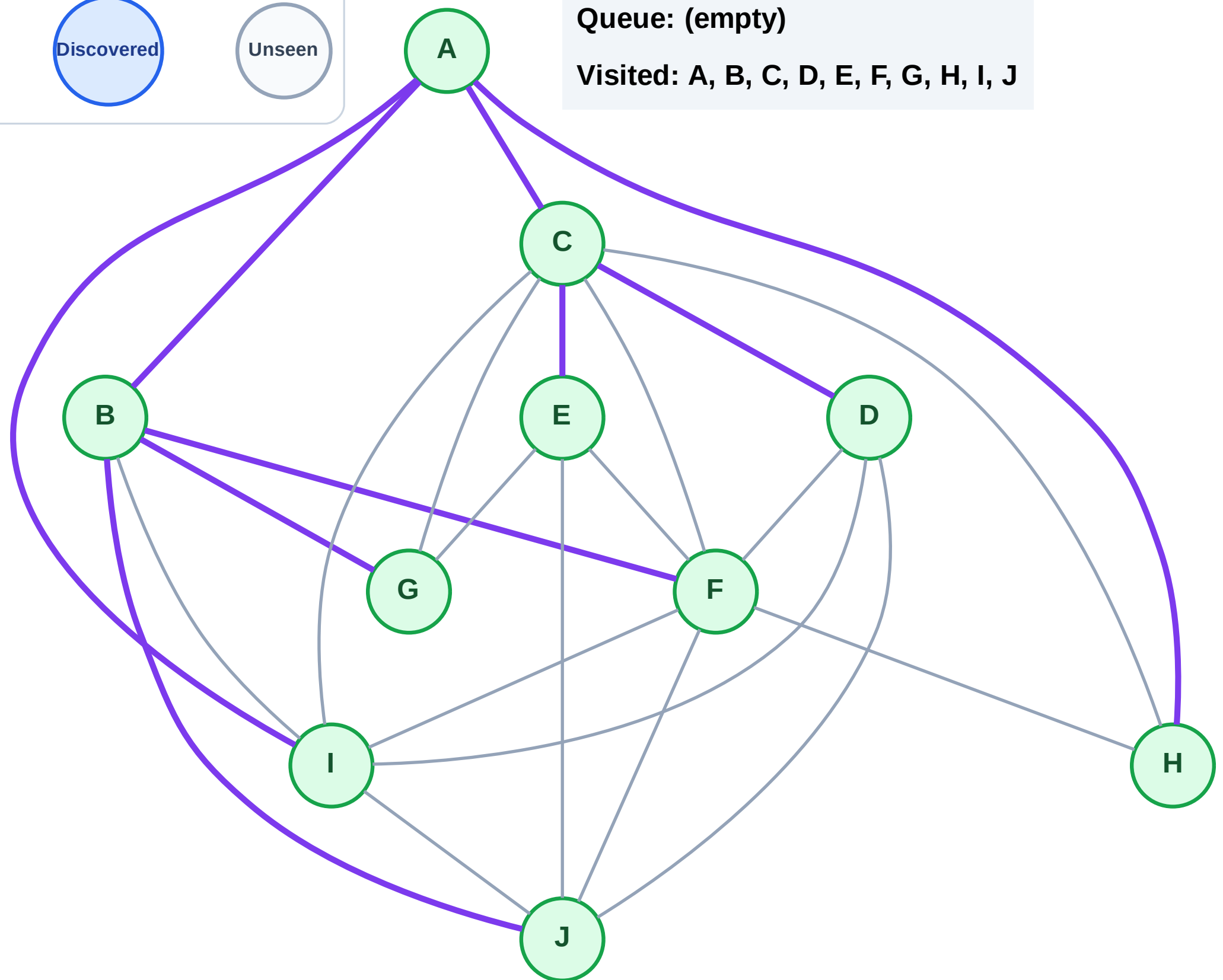
Step 21: No new neighbors from E

Legend



Queue: (empty)

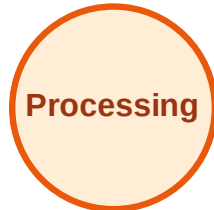
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

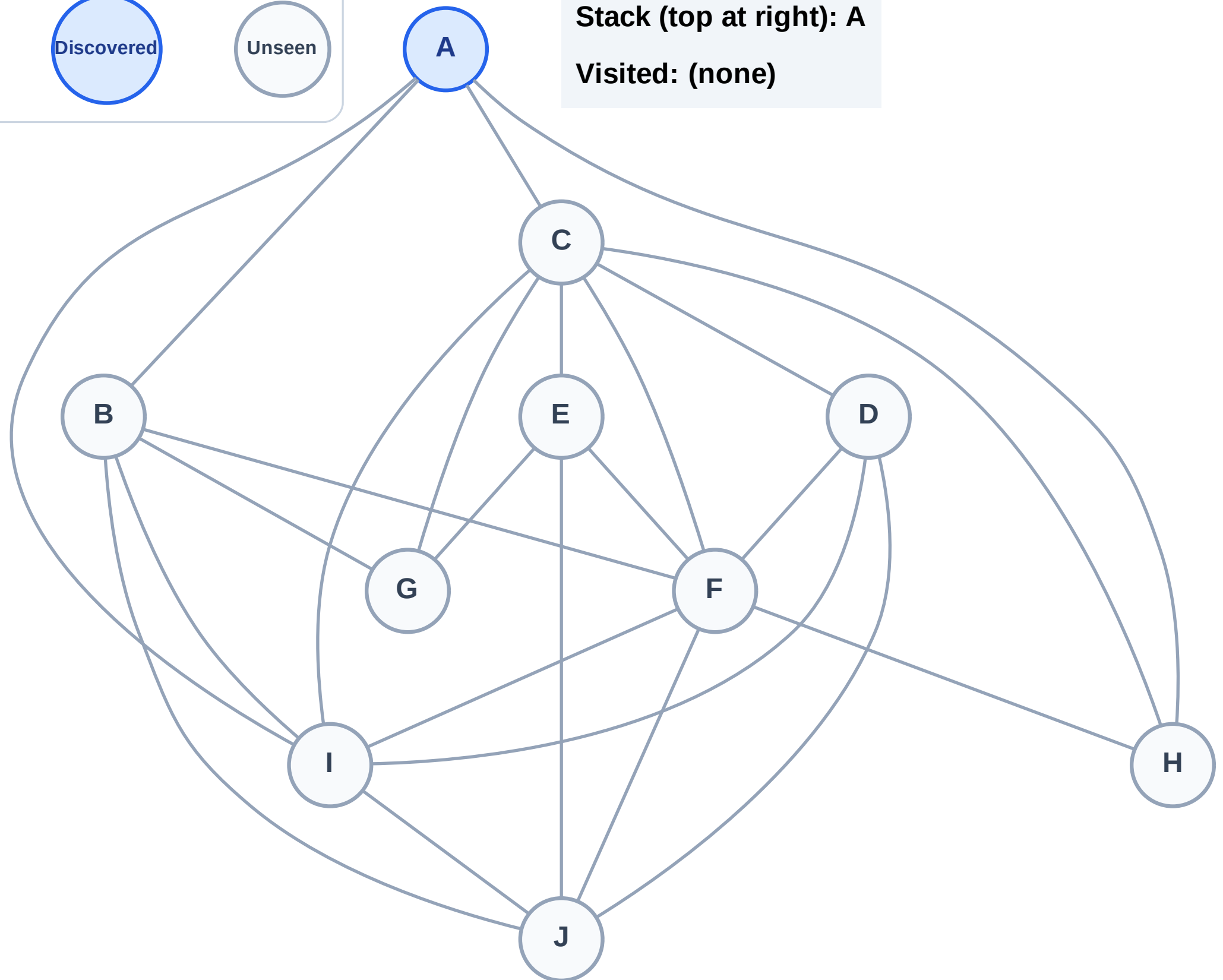
Step 1: Start at A

Legend



Stack (top at right): A

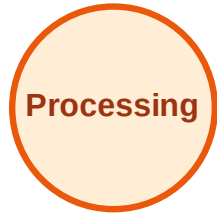
Visited: (none)



DFS Traversal

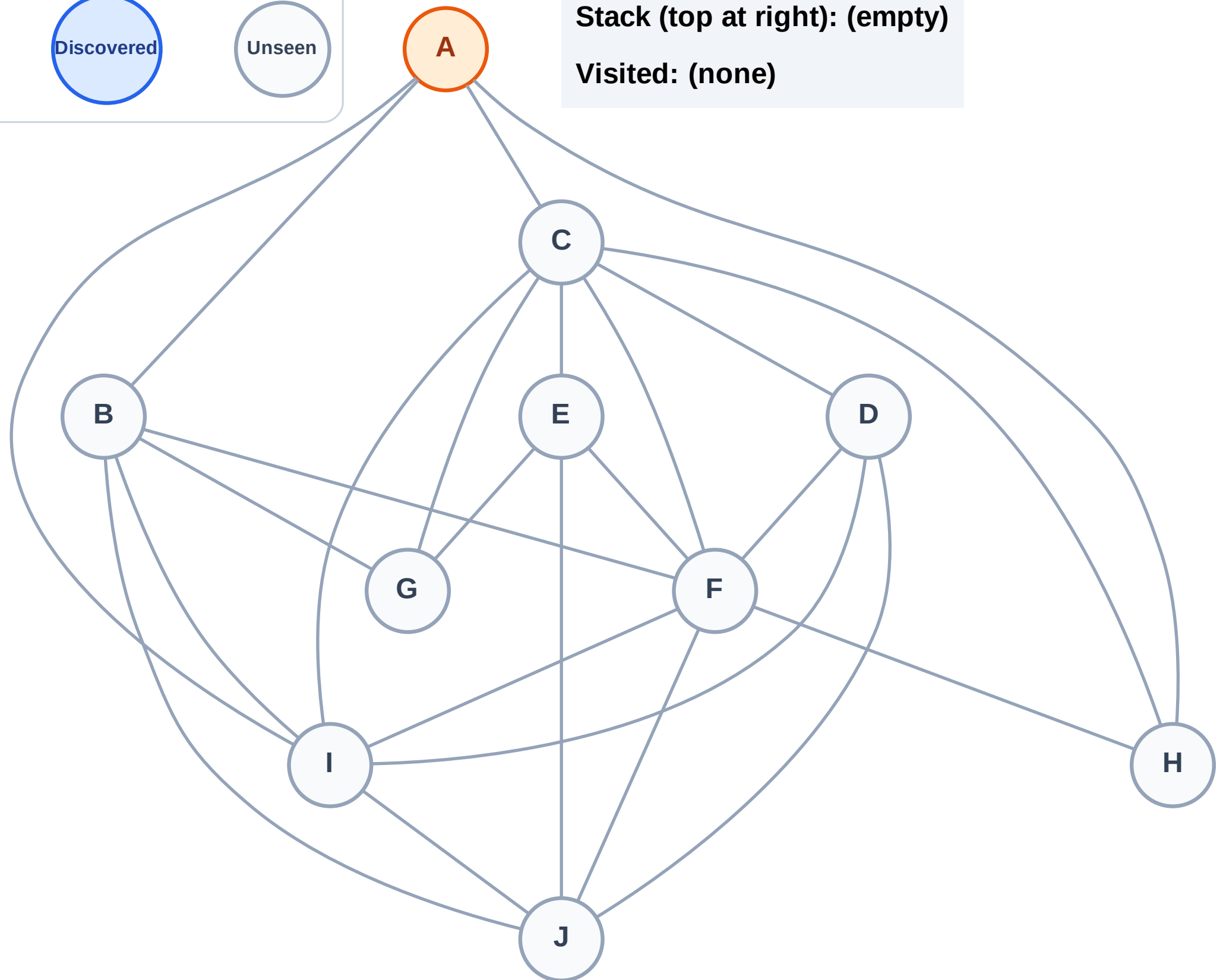
Step 2: Process node A

Legend



Stack (top at right): (empty)

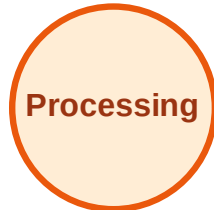
Visited: (none)



DFS Traversal

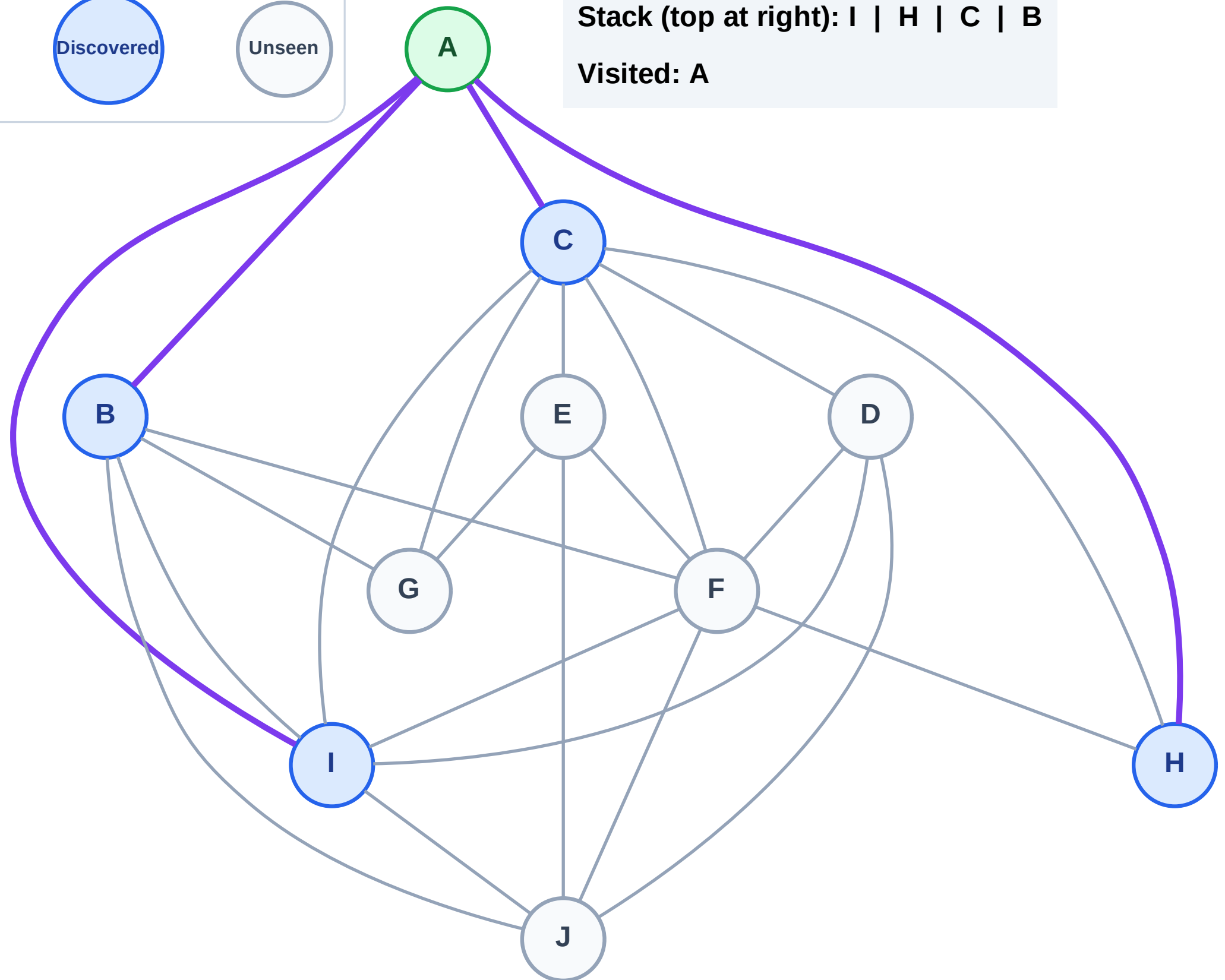
Step 3: Discover I, H, C, B from A

Legend



Stack (top at right): I | H | C | B

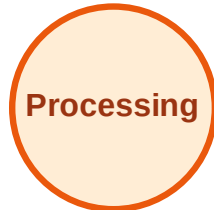
Visited: A



DFS Traversal

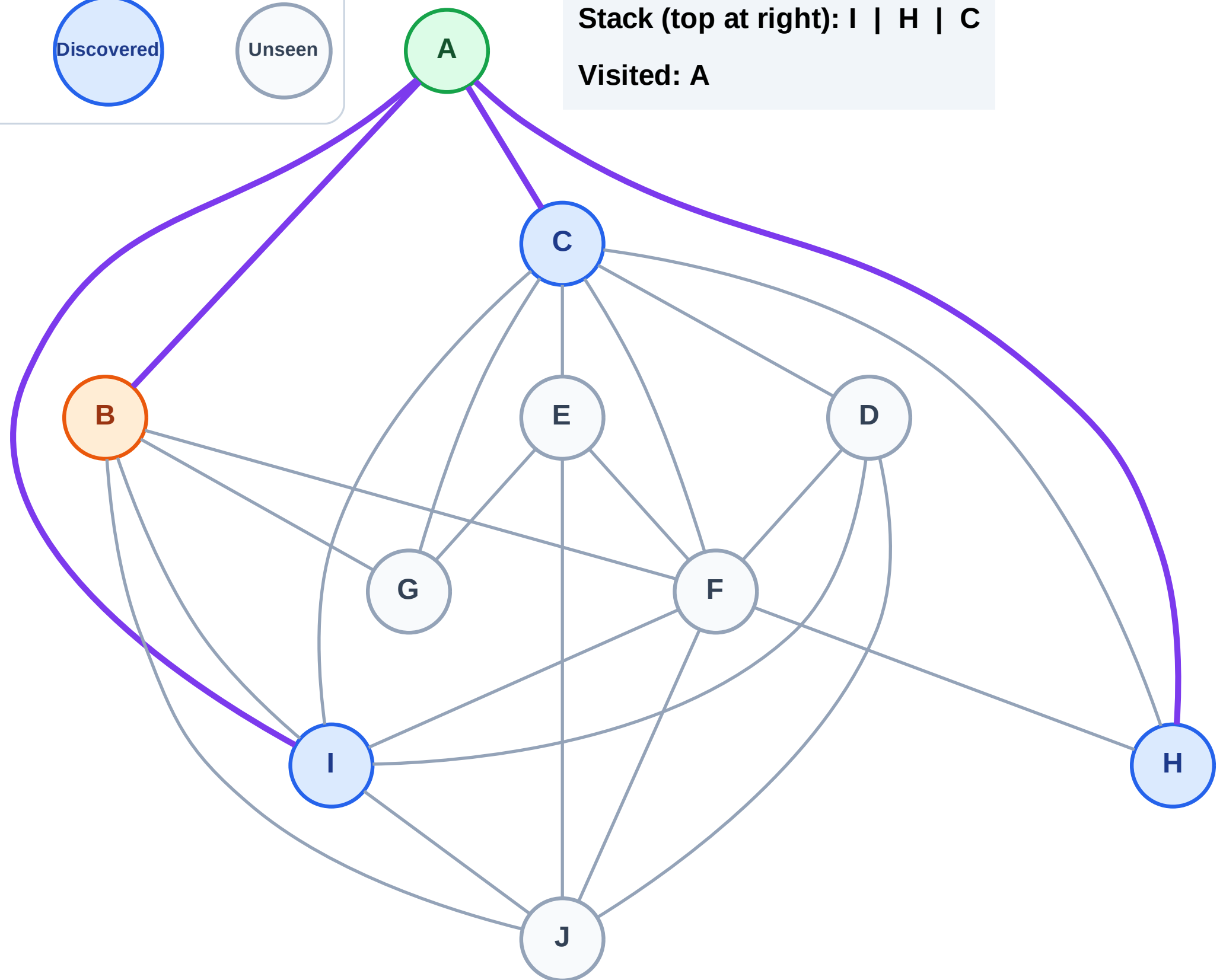
Step 4: Process node B

Legend



Stack (top at right): I | H | C

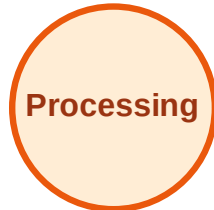
Visited: A



DFS Traversal

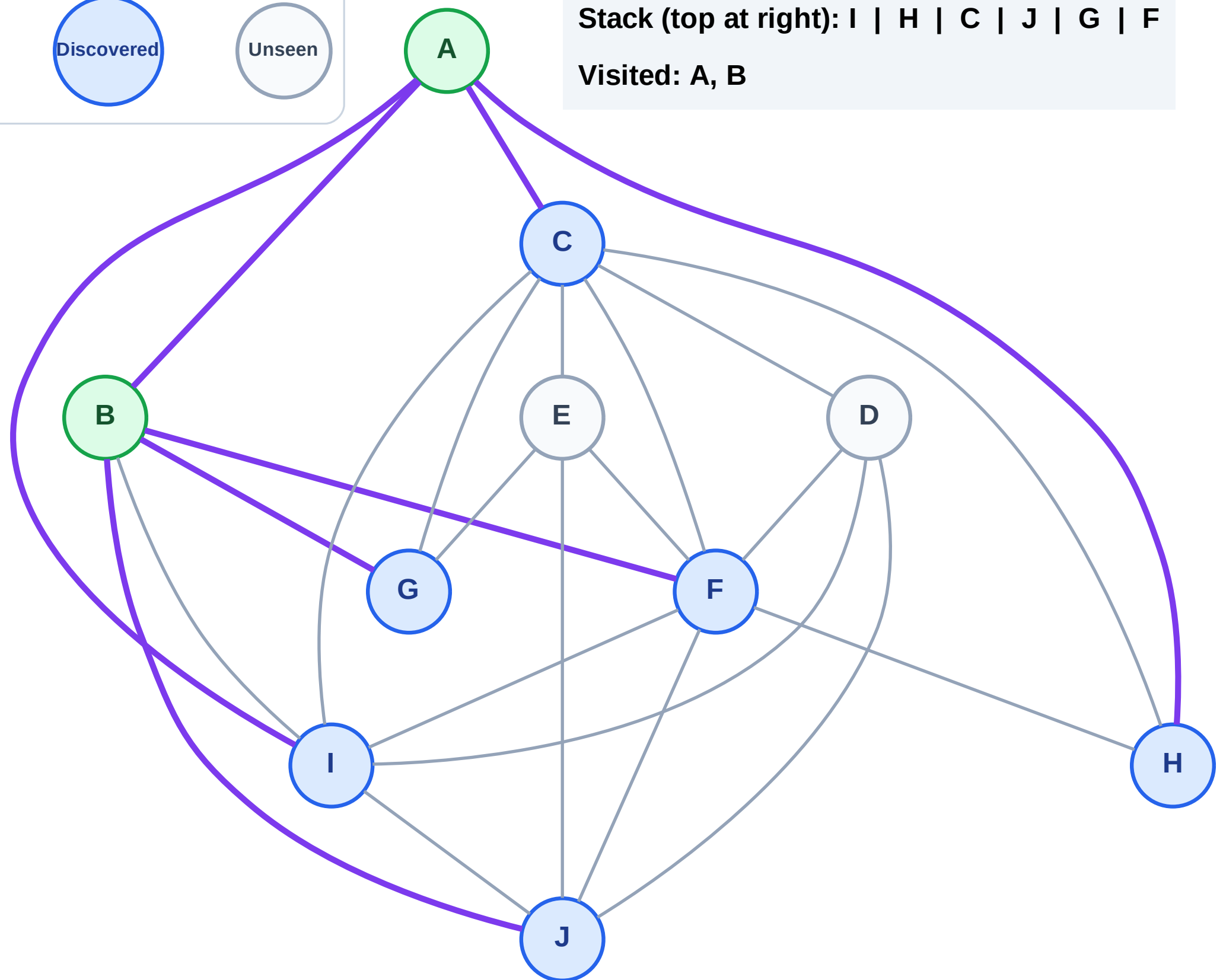
Step 5: Discover J, G, F from B

Legend



Stack (top at right): I | H | C | J | G | F

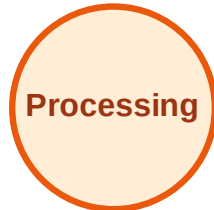
Visited: A, B



DFS Traversal

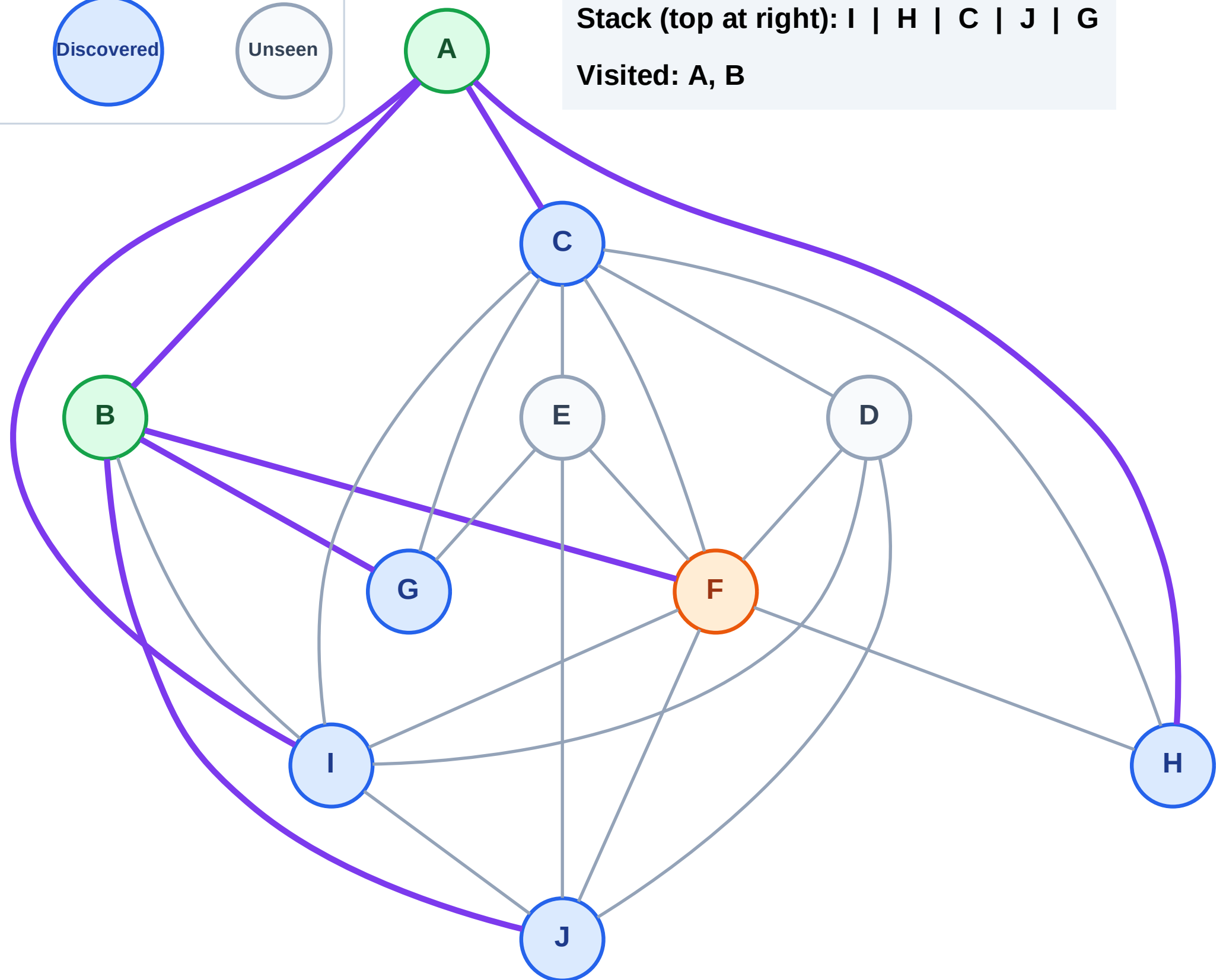
Step 6: Process node F

Legend



Stack (top at right): I | H | C | J | G

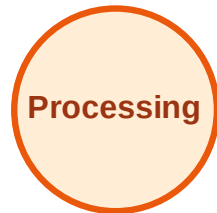
Visited: A, B



DFS Traversal

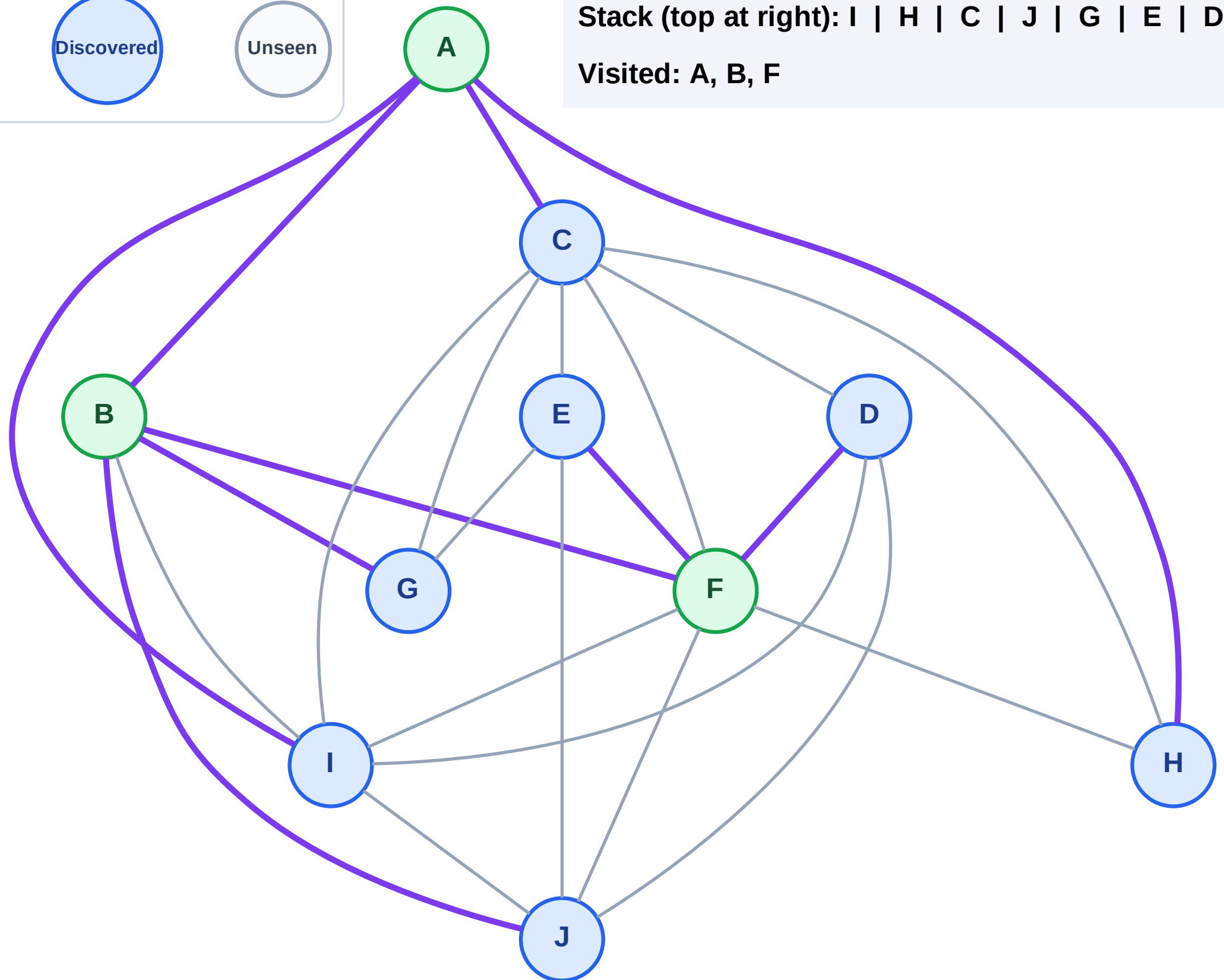
Step 7: Discover E, D from F

Legend



Stack (top at right): I | H | C | J | G | E | D

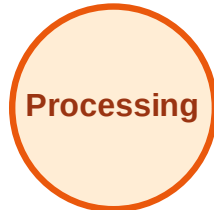
Visited: A, B, F



DFS Traversal

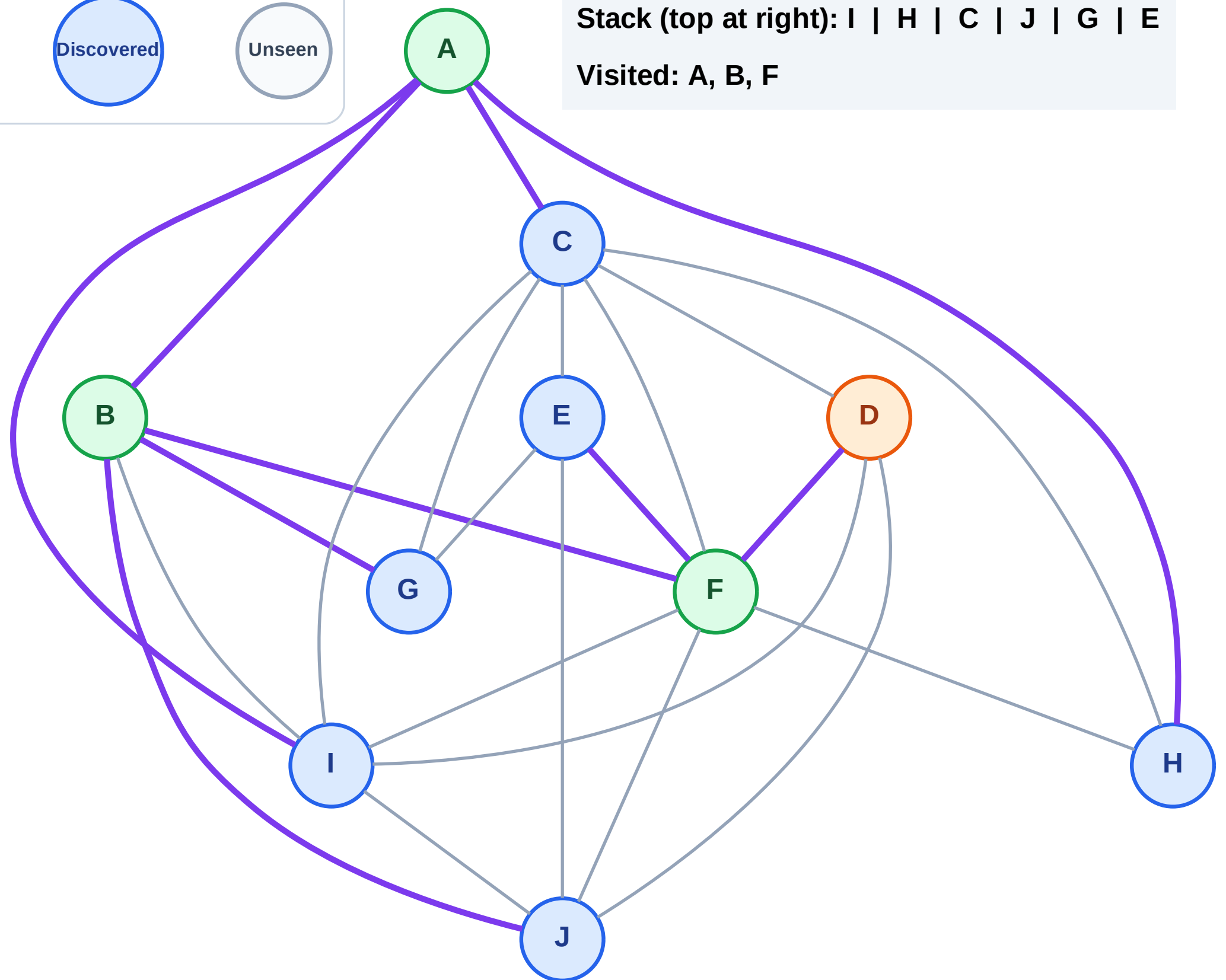
Step 8: Process node D

Legend



Stack (top at right): I | H | C | J | G | E

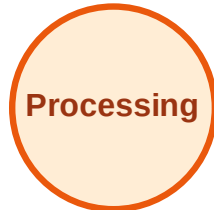
Visited: A, B, F



DFS Traversal

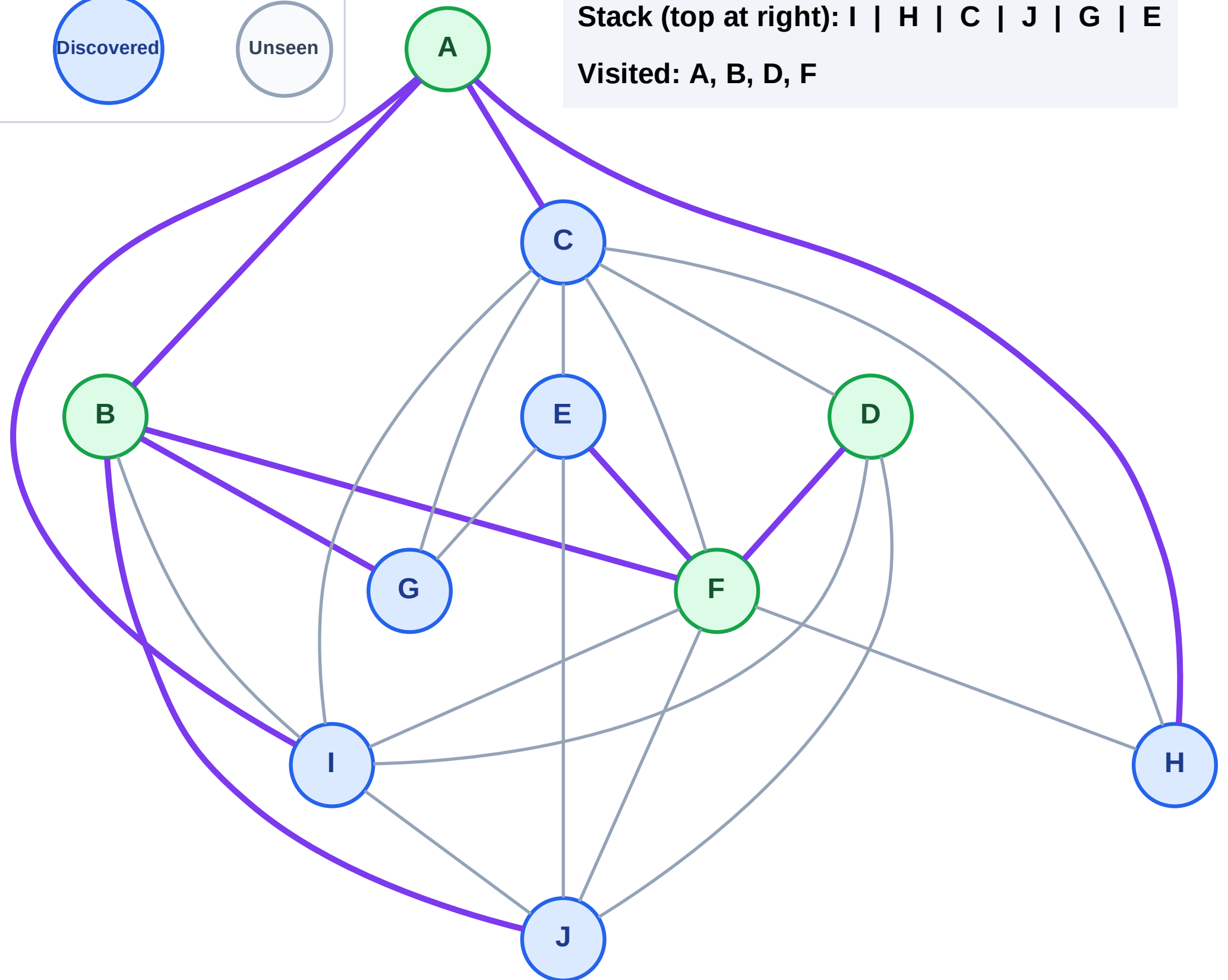
Step 9: No new neighbors from D

Legend



Stack (top at right): I | H | C | J | G | E

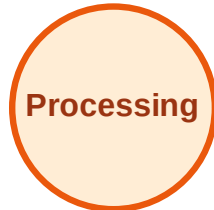
Visited: A, B, D, F



DFS Traversal

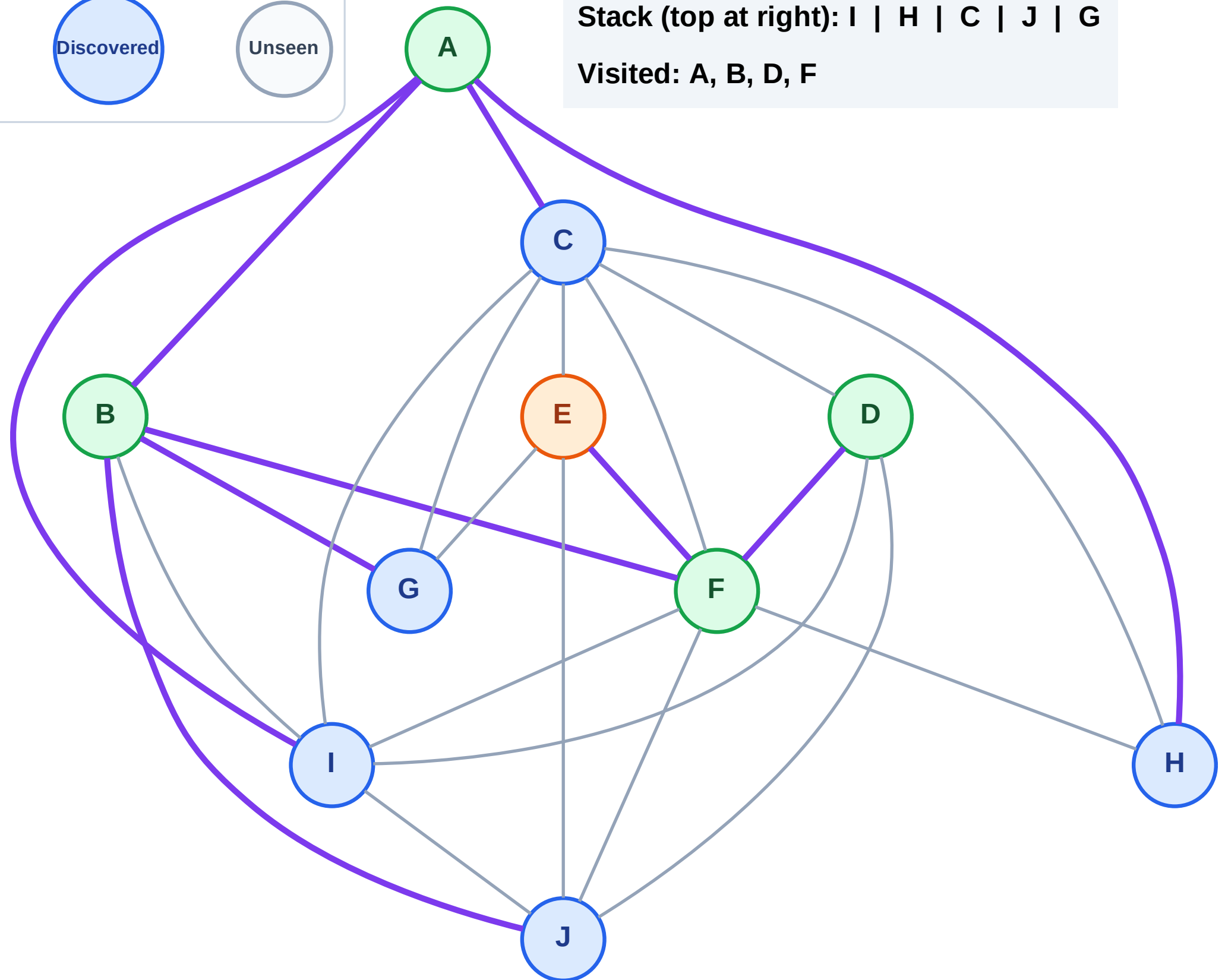
Step 10: Process node E

Legend



Stack (top at right): I | H | C | J | G

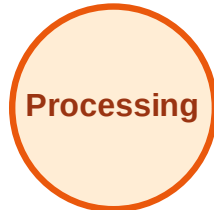
Visited: A, B, D, F



DFS Traversal

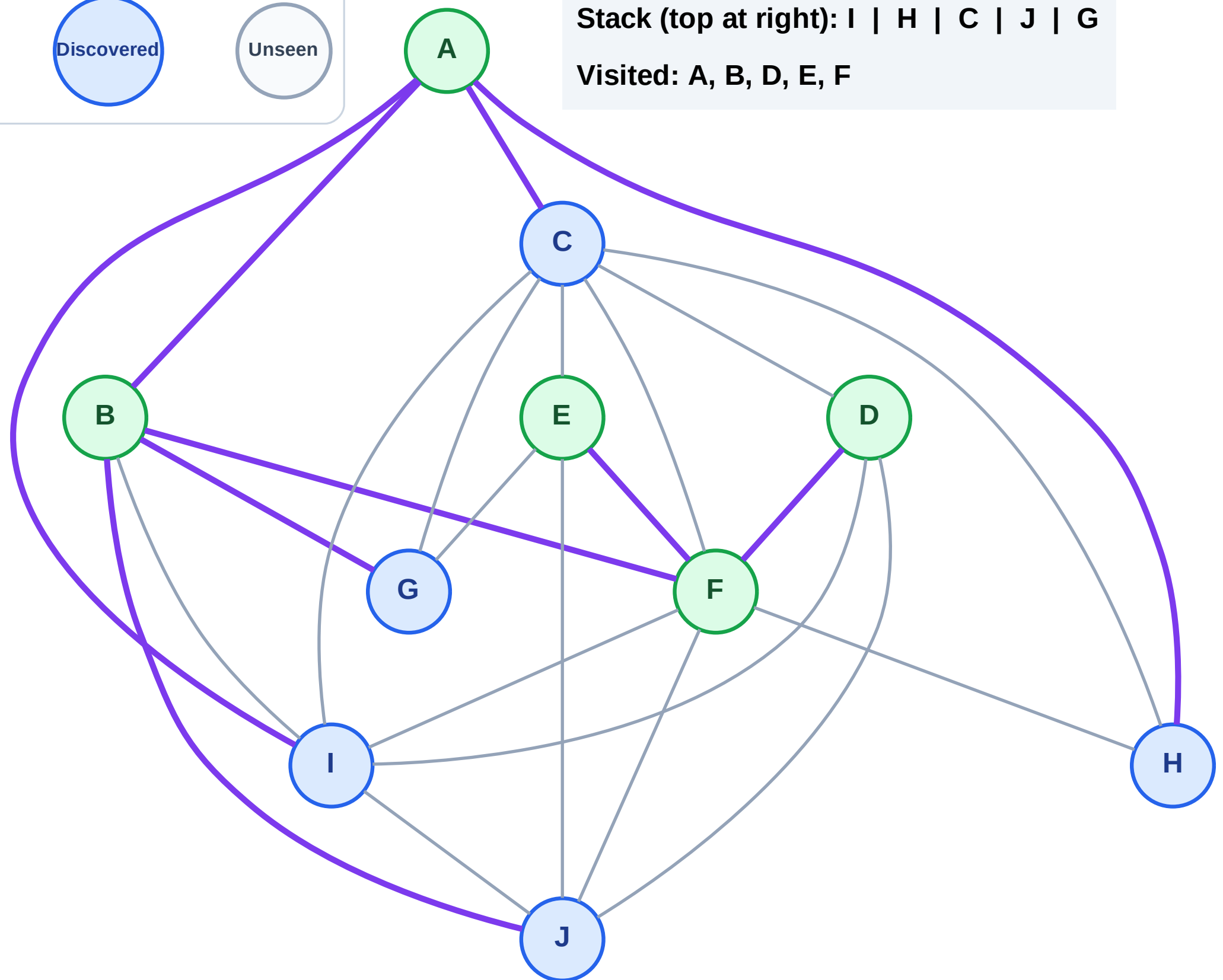
Step 11: No new neighbors from E

Legend



Stack (top at right): I | H | C | J | G

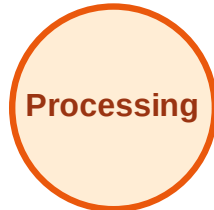
Visited: A, B, D, E, F



DFS Traversal

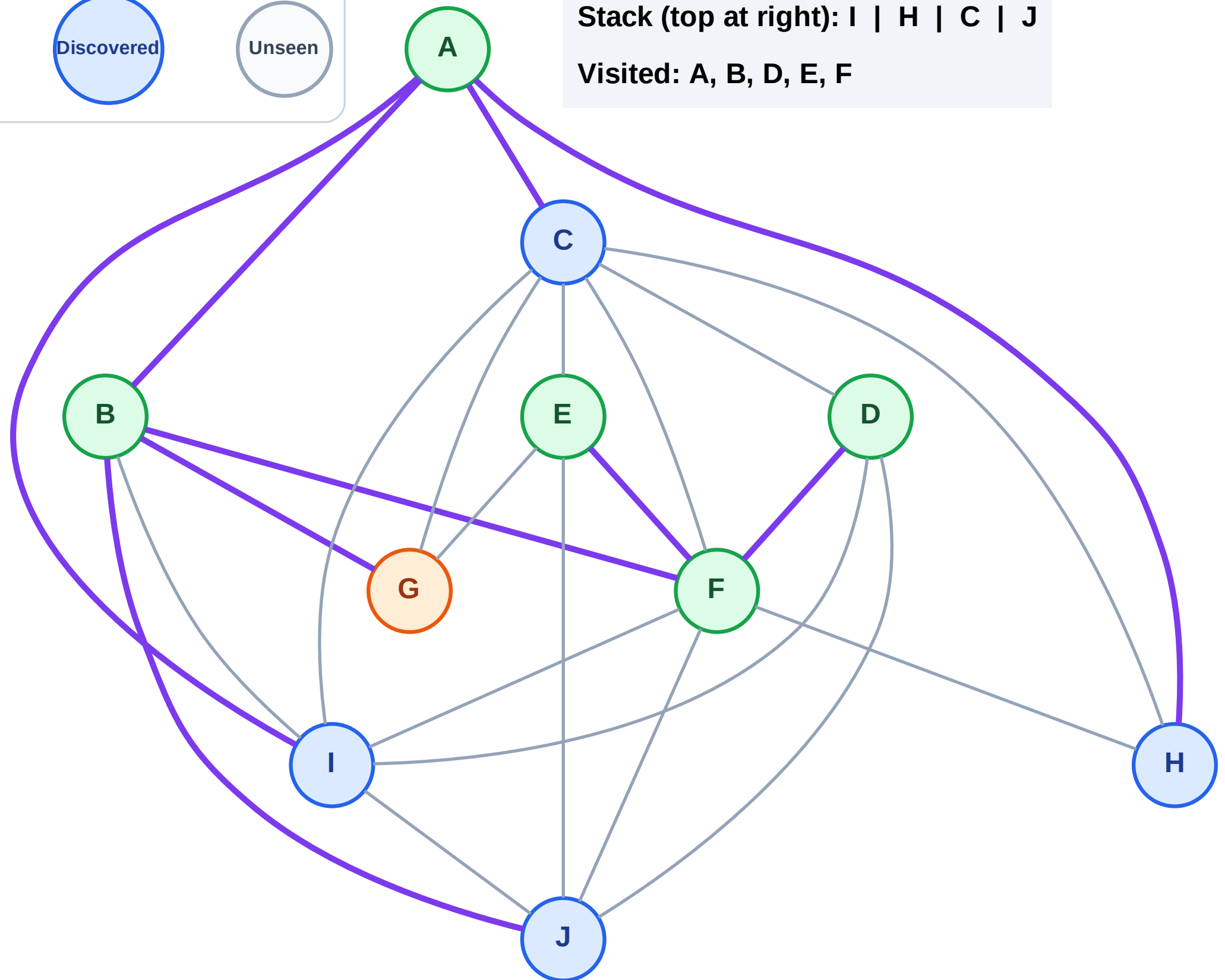
Step 12: Process node G

Legend



Stack (top at right): I | H | C | J

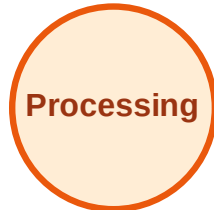
Visited: A, B, D, E, F



DFS Traversal

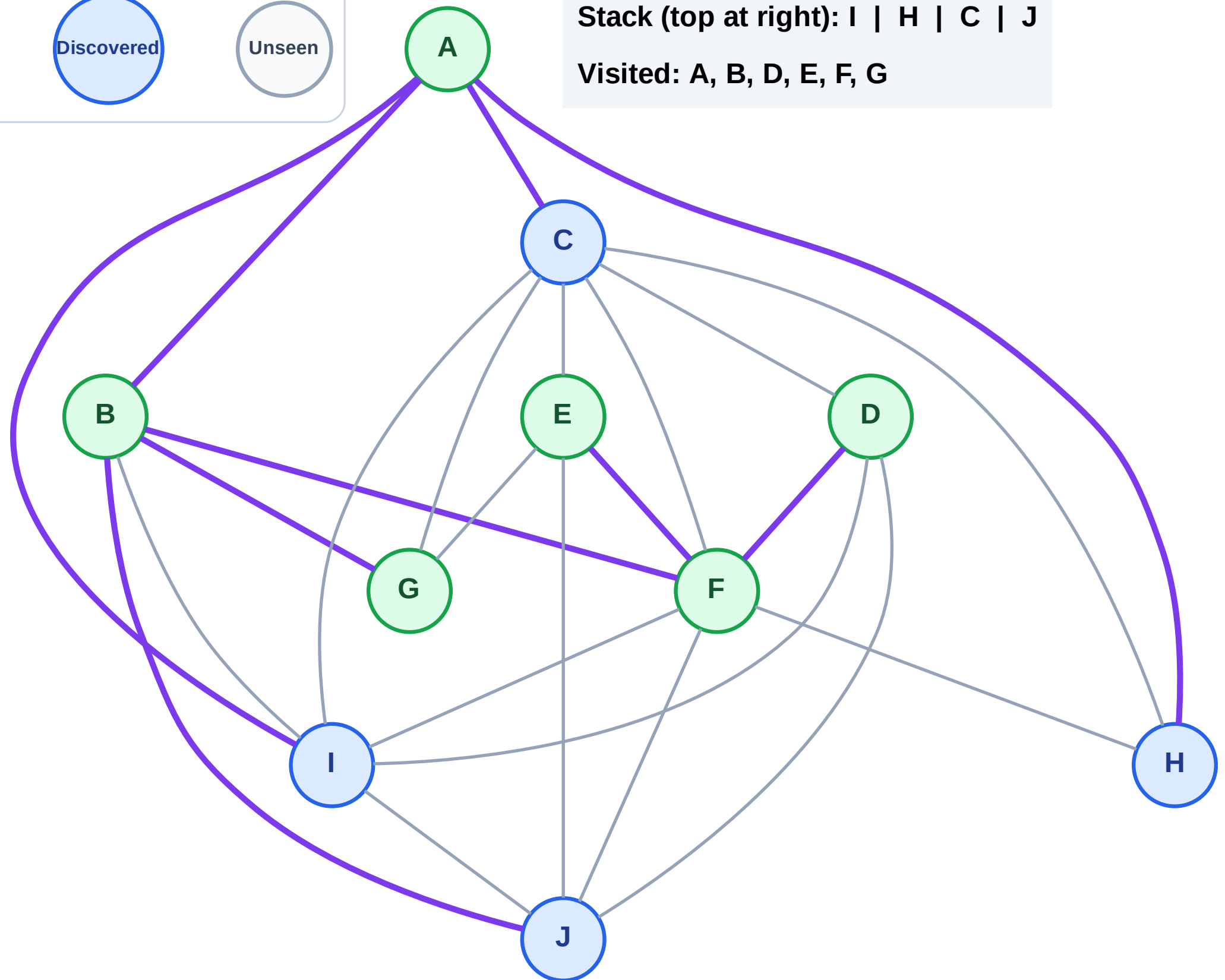
Step 13: No new neighbors from G

Legend



Stack (top at right): I | H | C | J

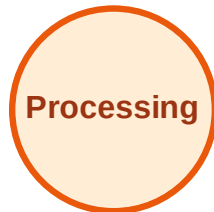
Visited: A, B, D, E, F, G



DFS Traversal

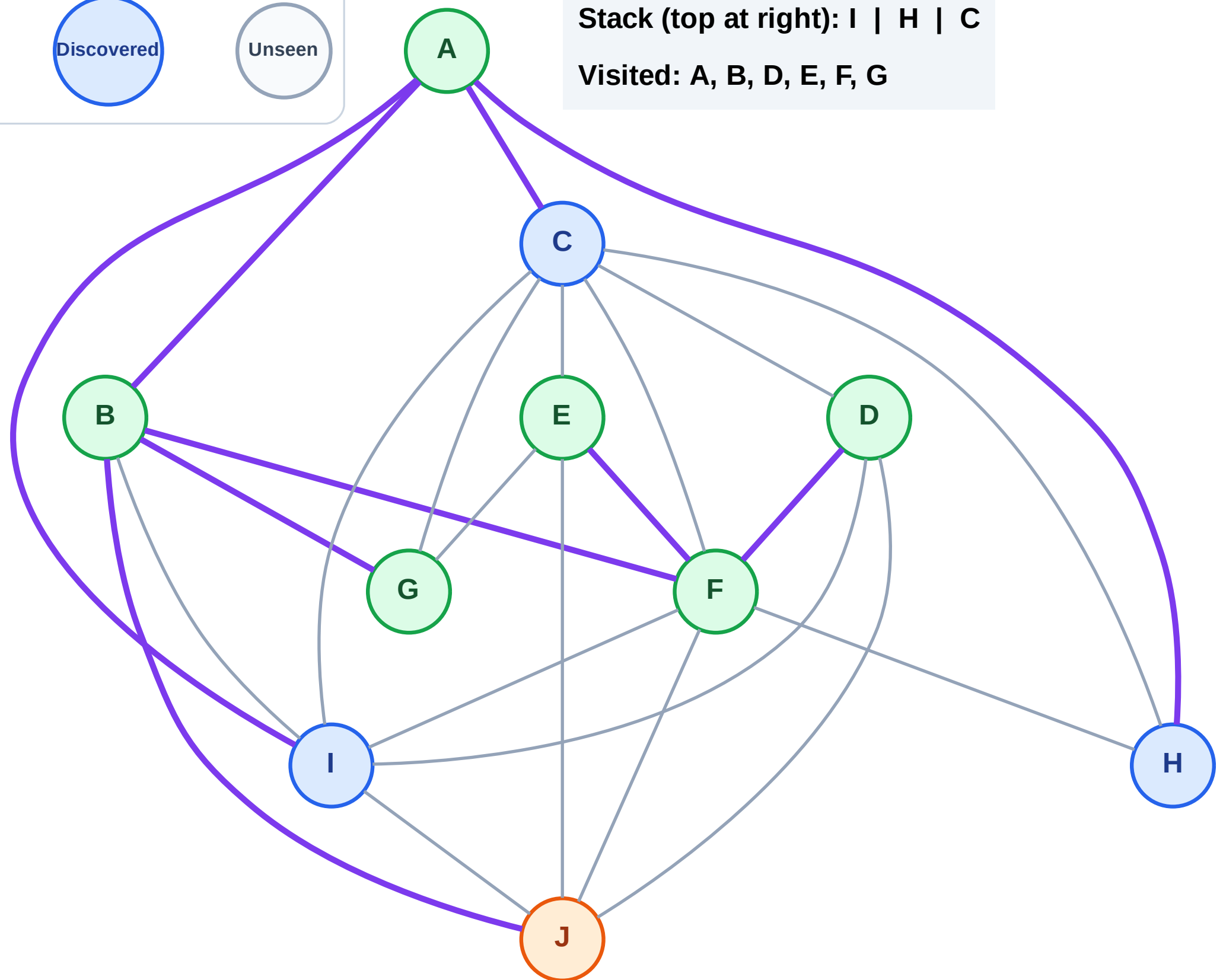
Step 14: Process node J

Legend



Stack (top at right): I | H | C

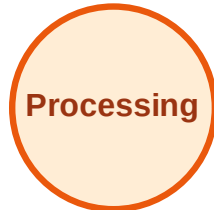
Visited: A, B, D, E, F, G



DFS Traversal

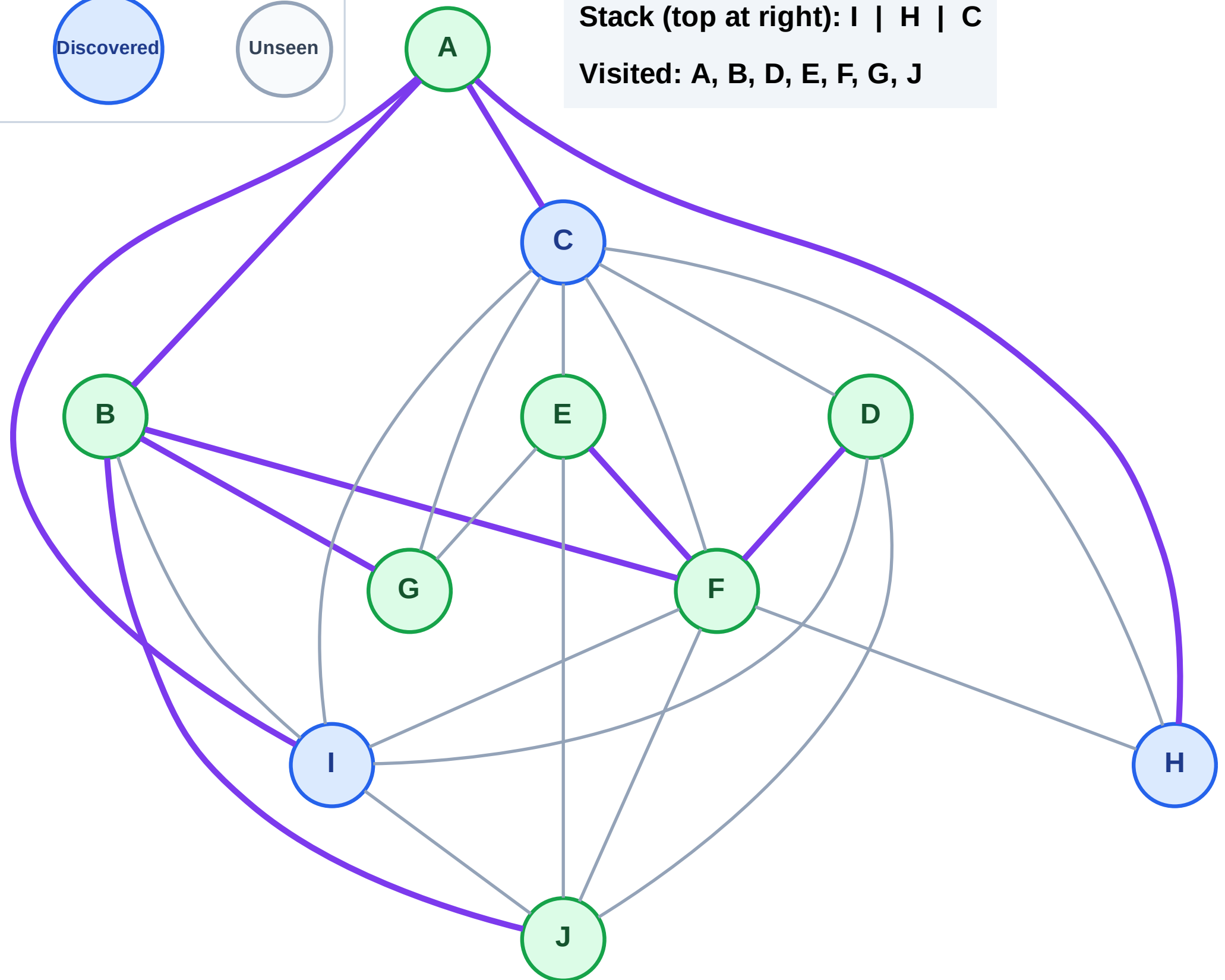
Step 15: No new neighbors from J

Legend



Stack (top at right): I | H | C

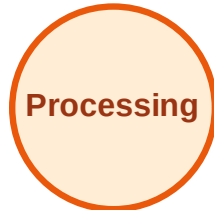
Visited: A, B, D, E, F, G, J



DFS Traversal

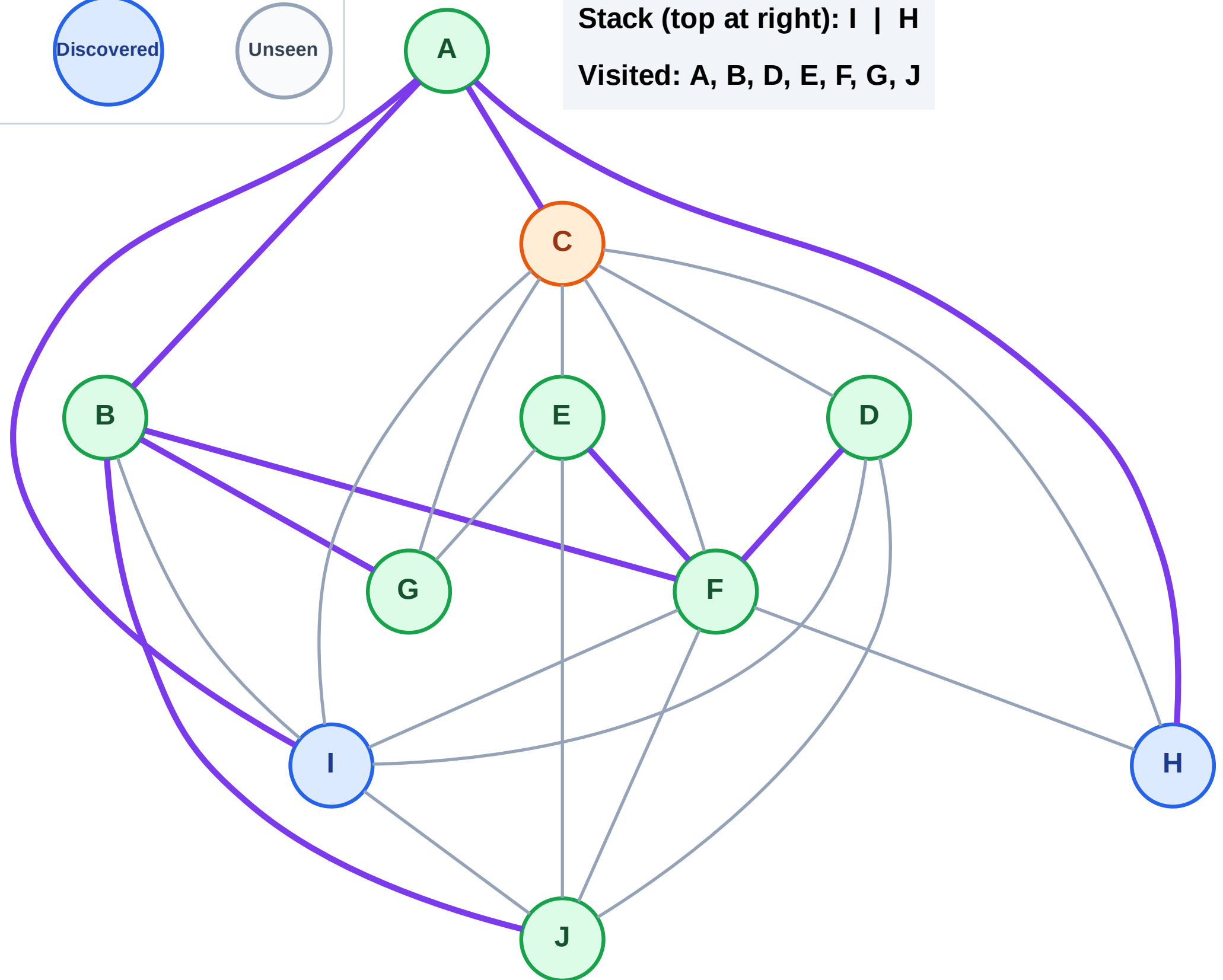
Step 16: Process node C

Legend



Stack (top at right): I | H

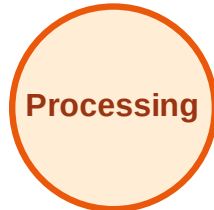
Visited: A, B, D, E, F, G, J



DFS Traversal

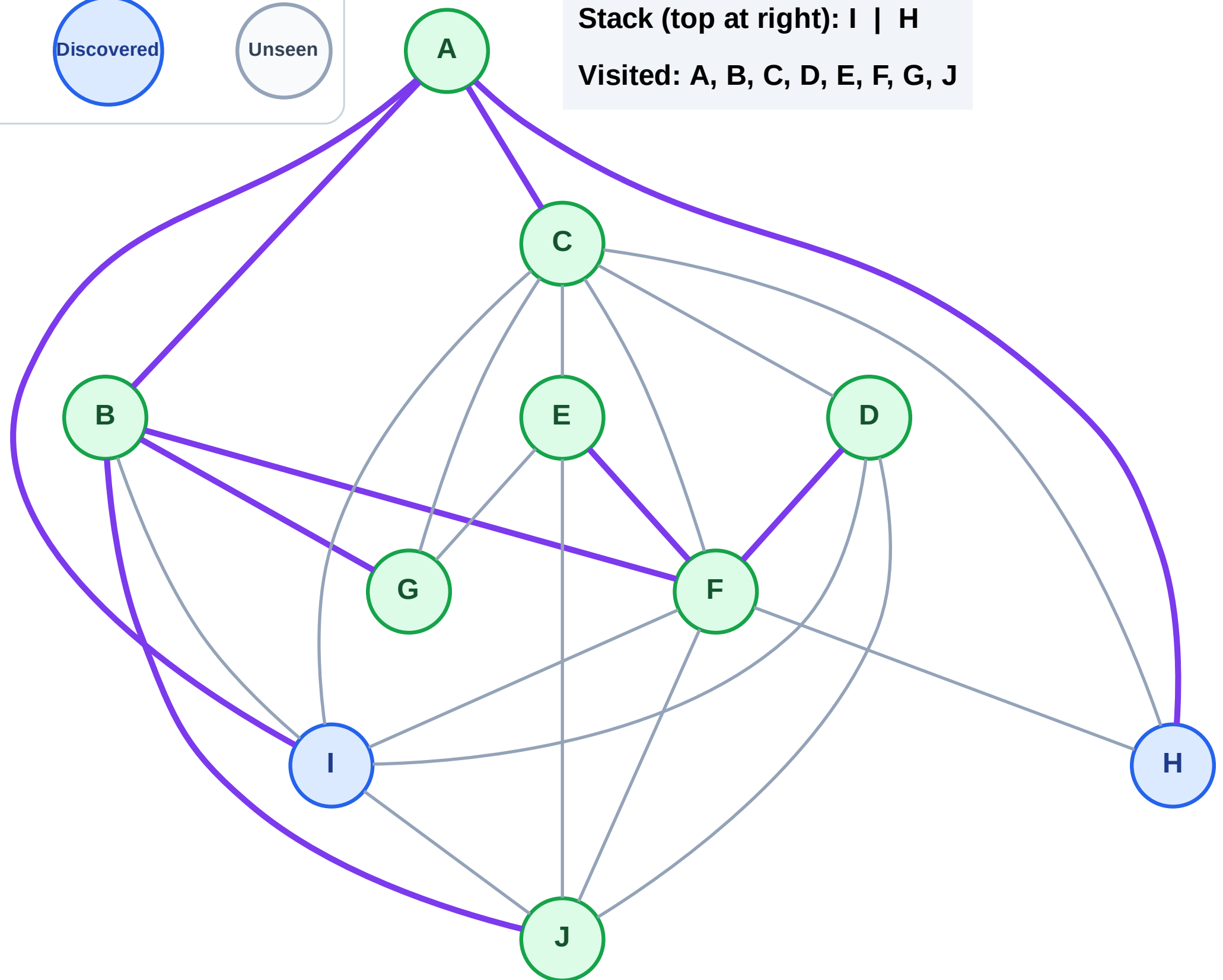
Step 17: No new neighbors from C

Legend



Stack (top at right): I | H

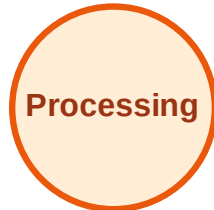
Visited: A, B, C, D, E, F, G, J



DFS Traversal

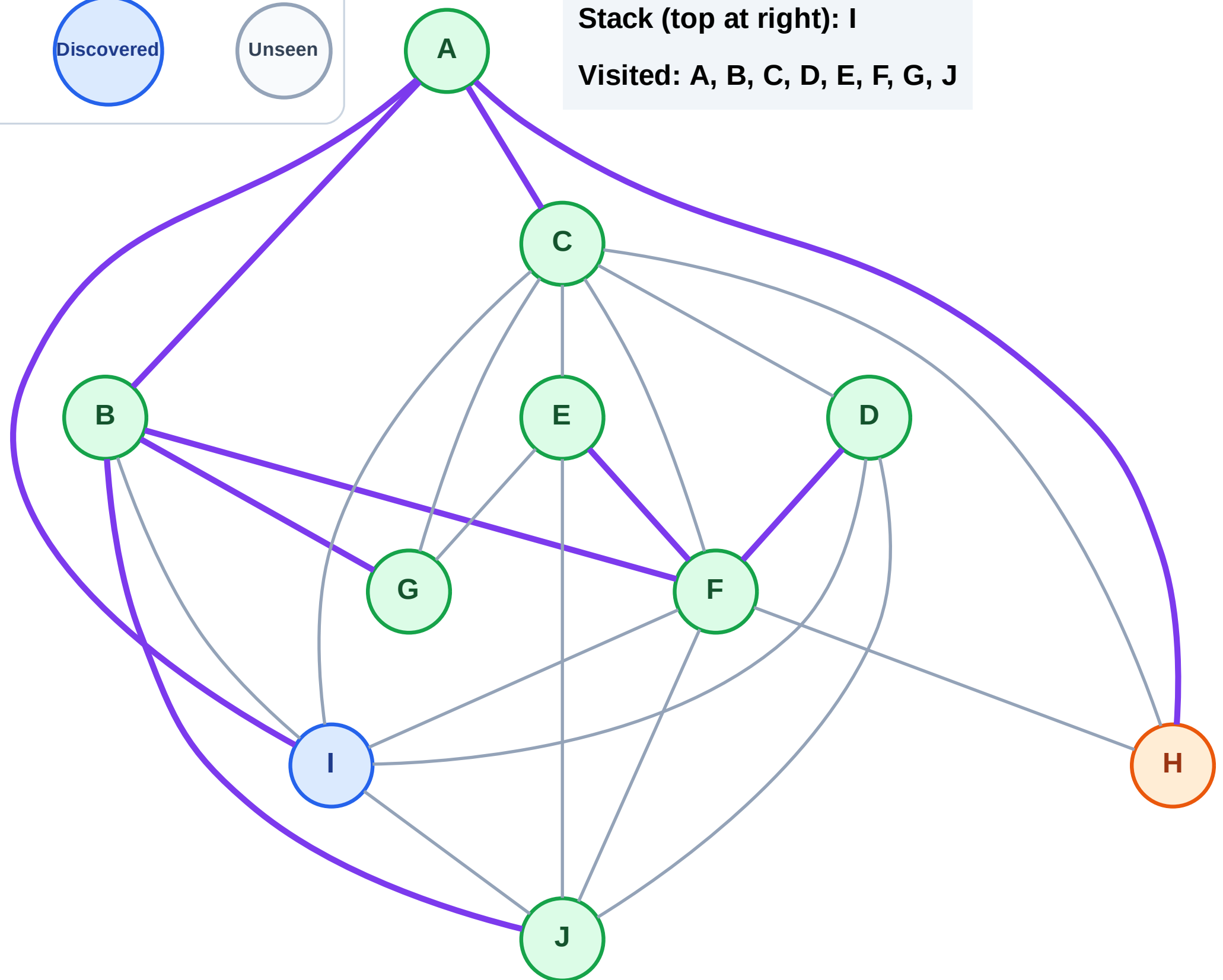
Step 18: Process node H

Legend



Stack (top at right): I

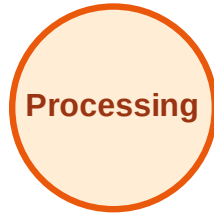
Visited: A, B, C, D, E, F, G, J



DFS Traversal

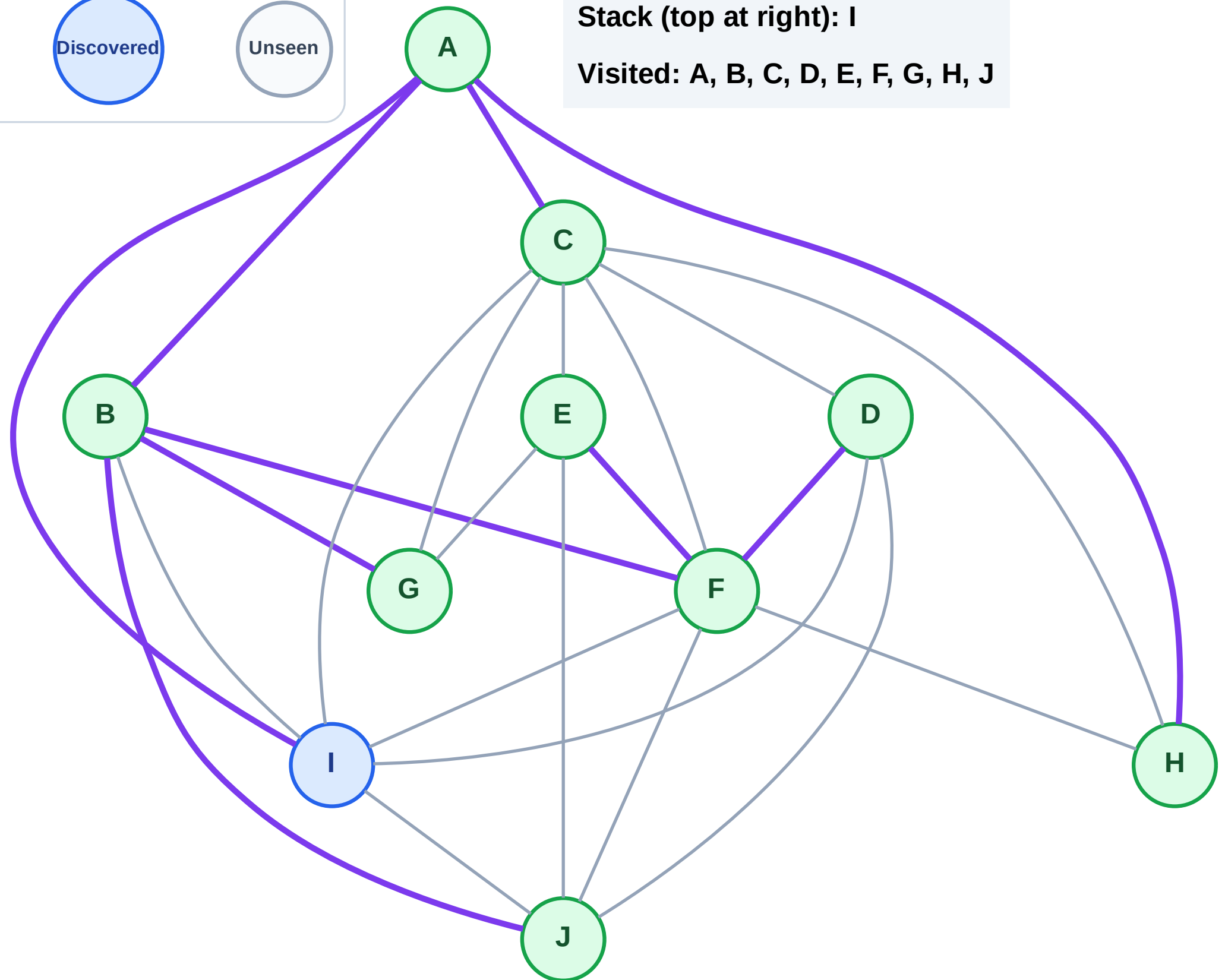
Step 19: No new neighbors from H

Legend



Stack (top at right): I

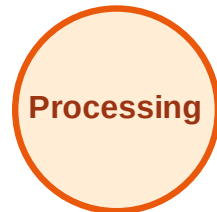
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

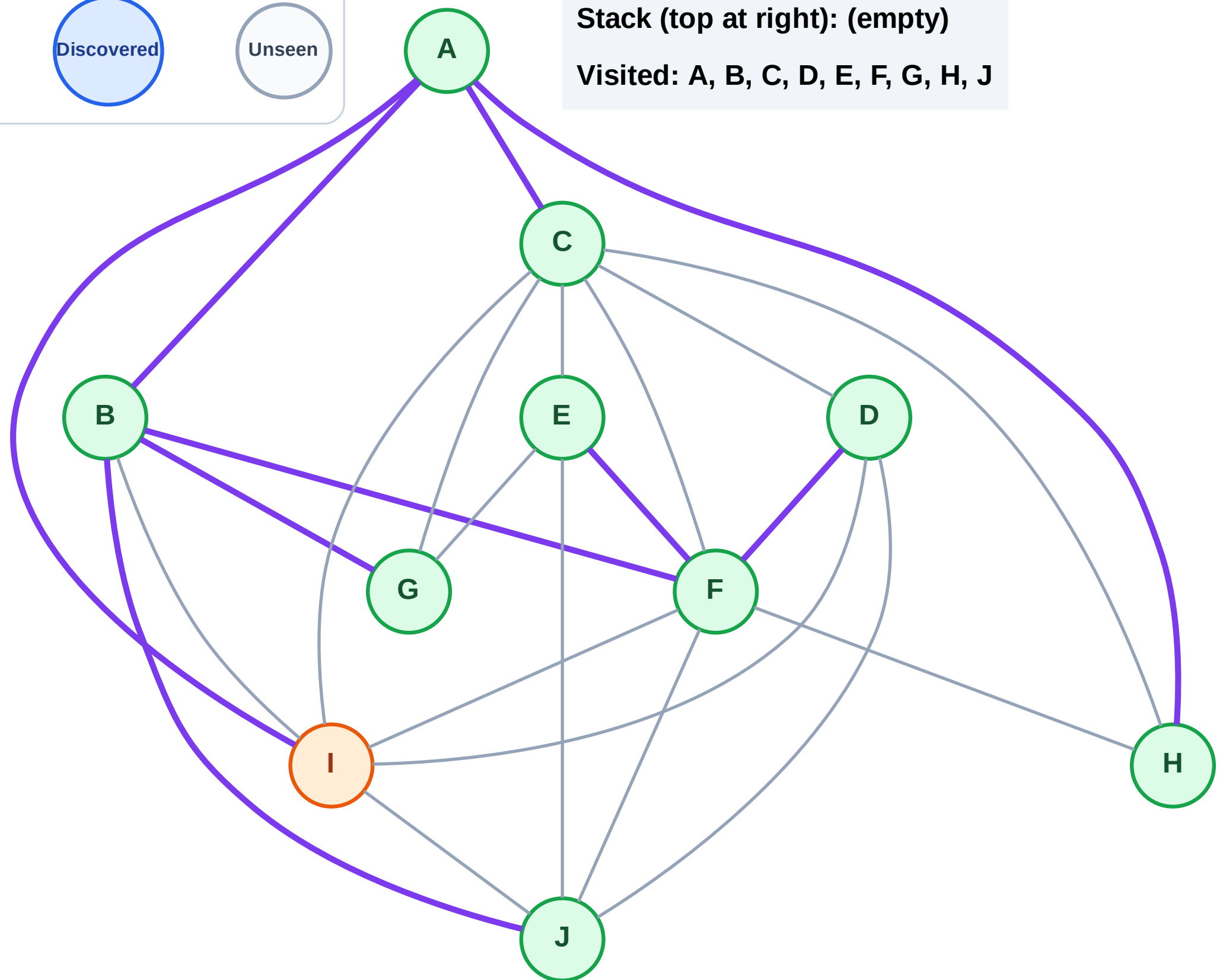
Step 20: Process node I

Legend



Stack (top at right): (empty)

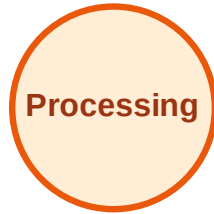
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

